



**University
of Victoria**

MECH 475 - Aircraft Design

Department of Mechanical Engineering

Final Project Report

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Executive Summary

This report presents the conceptual design of a vertical takeoff and landing (VTOL) aircraft optimized for emergency medical response in remote regions of British Columbia. The vehicle is intended to bridge the operational gap between conventional ambulances and full-scale rescue helicopters, offering rapid deployment from urban hospitals to inaccessible locations up to 100 km away.

Six configurations were evaluated using a weighted decision matrix based on noise, speed, emissions, footprint, and cost. A lift-and-cruise design with embedded ducted fans was selected, using a hybrid system with electric fans for vertical flight and a turboshaft-driven propeller for cruise. The final design has a maximum takeoff mass of 2453.61 kg, including a 500kg payload encompassing 4 passengers and approximately 100kg of medical equipment. The aircraft meets all performance requirements by balancing propulsion, wing loading, and the integration of the ducted fans into the wing structure.

The aircraft features a rectangular main wing with a span of 11 m, and a tapered tandem wing with a matching span. The main wing includes two ducted fans embedded in each half-span, while the tandem wing houses one fan per half-span. A novel shutter covering system is included over the fan ducts which closing during forward flight with a goal of reducing drag and increasing efficiency.

The design was iterated on to achieve stability coefficients within acceptable standard ranges. The final forward flight design point has a wing loading of 560.52 N/m² and a weight-to-power ratio of 0.0252 N/W. The vertical flight design point has a weight-to-disk-area ratio of 2093.98 N/m² and a weight-to-power ratio of 0.0076 N/W.

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1 Introduction

1.1 Motivation and Objectives

Emergency medical response times are critical, especially in urban regions where traditional ground or large-scale air ambulance services face serious limitations. In many areas of Northern British Columbia, difficult terrain and long travel distances create delays that can significantly impact patient outcomes. This project is motivated by the need to enhance emergency response capabilities beyond dense urban centers. The primary objective is to design a conceptual VTOL aircraft capable of rapidly deploying from an urban hospital hub to remote locations up to 100 km away, ensuring a 200 km round-trip range. The aircraft must safely transport a stretchered patient, a paramedic, and essential medical equipment, while being able to operate in areas without runways through vertical takeoff and landing. It will leverage both electric and combustion propulsions systems to optimize vertical lift and forward cruise performance and remain compact enough to use in tight rescue missions and populated areas without infrastructure. By combining these design criteria into a single vehicle concept, the aircraft aims to fill the gap between ambulances and specialized rescue helicopters.

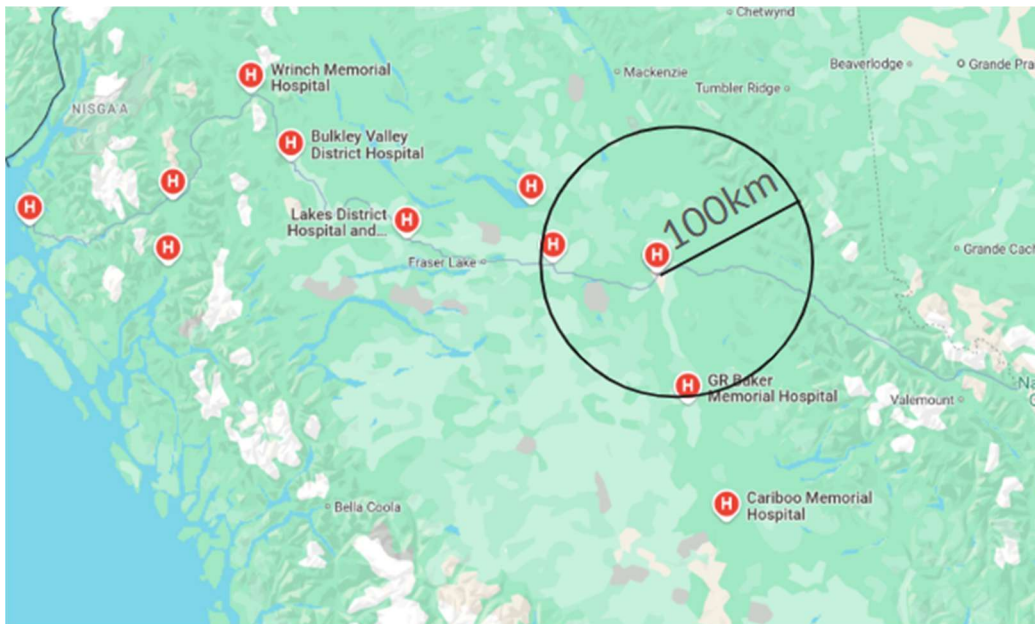


Figure 1 - Accessible area around Prince George

1.2 Design Requirements

To meet the goals of rapid emergency response and regional medical accessibility, the aircraft must satisfy a defined set of performance and operational criteria. It must be capable of vertical takeoff and landing from confined areas, while maintaining safe and stable forward flight

at a cruise speed of approximately 60 m/s. The design must support a total range of 200 km round trip from the hospital to a remote location without the need for refueling.

The cabin must accommodate a stretchered patient, a paramedic, a pilot, and an additional passenger along with onboard emergency equipment. The propulsion system must sustain both vertical and horizontal flight modes, with sufficient redundancy to allow safe mission completion in the event of partial system failure. While noise is not a primary concern due to the emergency nature of the aircraft, emission should be minimized using a hybrid-electric system. Additionally, the overall footprint of the aircraft will be minimized to support operating in confined landing zones.

Finally, the aircraft must meet general aviation stability and control standards, including appropriate static margin and center of gravity positioning, to ensure safety across all phases of flight. These requirements advised key design decisions in sizing, configuration, propulsion, and aerodynamic shaping.

1.3 Market Analysis and State-of-the-Art

A market study was conducted to assess current transportation solutions and compare their respective pros and cons. Each vehicle was chosen as they are either actively covering the role of emergency vehicle or would be the best option available.



Figure 2 - City Ambulance

City ambulances are equipped to transport one patient in a stretcher with up to two EMTs and a driver [1]. These vehicles are also equipped with gear capable of delivering first aid, stabilizing patients and monitoring vitals. Their pros include fast urban response; national average of nine-minute response; a patient in the stretcher with an additional passenger sitting on a bench; can administer treatment during transport; cost effective and suitable for city infrastructure. Some of the cons include being dependent on existing city infrastructure, road conditions and is potentially dangerous for staff and patients at high speeds.



Figure 3 - Sikorsky S76 Air Ambulance

The Sikorsky S76 Air Ambulance is a specialized in remote or longer ranged transport of patients where it is crucial for the patient to be minimally disturbed [2]. Their pros include fast long-range transport, access to remote areas, can carry two patients and up to four medical personnel. Some cons include being expensive to operate, limited use in urban environments requiring landing pads, and they are impractical for short to medium distances.



Figure 4 - CH-149 Cormorant Rescue Helicopter

The CH-149 Cormorant is a Long-range helicopter equipped with state-of-the-art technology specifically dedicated to search and rescue operations [3]. Its pros include very long range and high speed, very high capacity for patients and cargo, and can operate under harsh

weather conditions. Its cons include limited number in operation, very high cost, and limited to unusable in urban environments.



Figure 5 - Bell Boeing V-22 Osprey

Lastly, outside of current medical based vehicles, a comparison to something based purely on transportation such as the Bell Boeing V-22 Osprey [4] is made. The Bell Boeing V-22 Osprey is a Tiltrotor aircraft designed for vertical takeoff and long-range high-speed cruise, designed with military and multipurpose application. When considering it for medical transport its pros include extreme crew and cargo capacity, extreme range and speed and has uses outside of medical application. Its cons include being prone to engine failure with minimal safety measures, size makes it impractical for urban and tight areas and is extremely costly.

To conclude this market study, we found the challenges of current emergency response vehicles are summarized as:

- Specialized equipment and pilots - Lack of personnel
- Poor weather conditions
- Drones and birds for low flying
- Flight crew fatigue
- Speed reduction in urban areas
- High operation cost (fuel and personnel)

1.4 Mission Profile

The mission profile for this aircraft is based on a round-trip rescue operation between an urban hospital in Prince George and remote areas within a 100 km radius, totaling a range of 200 km as shown in Figure 1. The flight sequence is divided into 17 stages, including vertical takeoff, transition, forward climb, cruise, loiter, descent, hover, and patient pickup.

The aircraft must perform two flight segments: one to the rescue site and one returning to the hospital, each involving altitude changed between ground and 1500 meters. Hovering near ground level upon arrival to the incident site may be required for specific missions, with loiter time allocated for site identification. The complete mission profile can be seen in Figure 6 with the segment labels in Table 1.

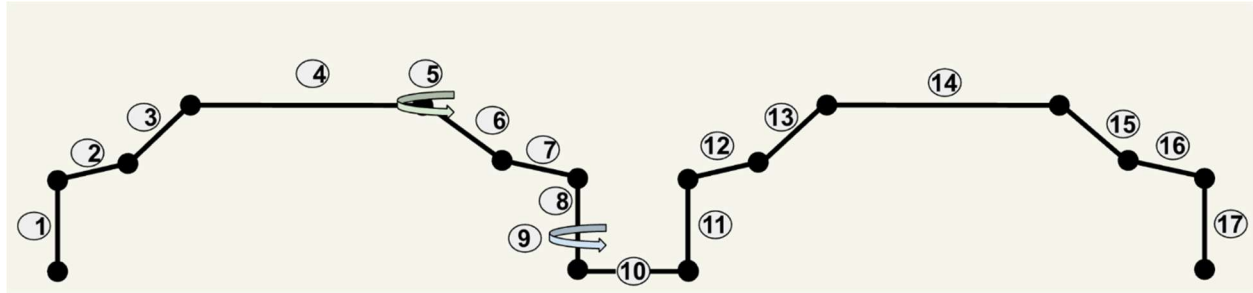


Figure 6 - Mission Profile

Table 1 - Mission profile

Stage	Description	Approx. Altitude	Stage	Description	Approx. Altitude
1	Vertical Climb	0-300m	10	Pick-up	0m
2	Transition	300 - 310m	11	Vertical Climb	0 - 300m
3	Forward Climb	310 - 1500m	12	Transition	300 - 310m
4	Cruise	1500m	13	Forward Climb	310 - 1500m
5	Loiter	1500m	14	Cruise	1500m
6	Descent	1500 - 310m	15	Descent	1500 - 310m
7	Transition	310 - 300m	16	Transition	310 - 300m
8	Vertical Descent	300 - 0m	17	Vertical Descent	300 - 0m
9	Hover	10 - 1m			

2 Concept Generation and Selection

In this section of the report, six initial concepts are explored and compared using the analytic hierarchy process (AHP). Each concept focuses on a unique flight configuration to offer

a diverse set of VTOL aircrafts that can fulfill the mission statement while being competitive designs against current real-world options. All vehicles in their preliminary design phase offer a unique VTOL mechanism, promote competition and demonstrate design concepts that offer a viable solution. In section 2.1, each design is presented with its respective benefits and limitations. In section 2.2, the concepts are weighted against each other, and the final design is selected.

2.1 Initial Concepts

The following concepts include of quadcopter designs with tilting and stationary rotors, fixed wing examples with lift propellers embedded in the wing or at the tip of the wing with a tilt mechanism. Lastly, two designs feature a main rotor like a helicopter or a gyroplane. This main rotor is designed to provide lift for VTOL flight conditions and offer a safe emergency landing procedure called autorotation, helicopter or a gyroplane. This main rotor is designed to provide lift for VTOL flight conditions and offer a safe emergency landing procedure called autorotation.

2.1.1 Design 1 - Tilt Rotor Quadcopter

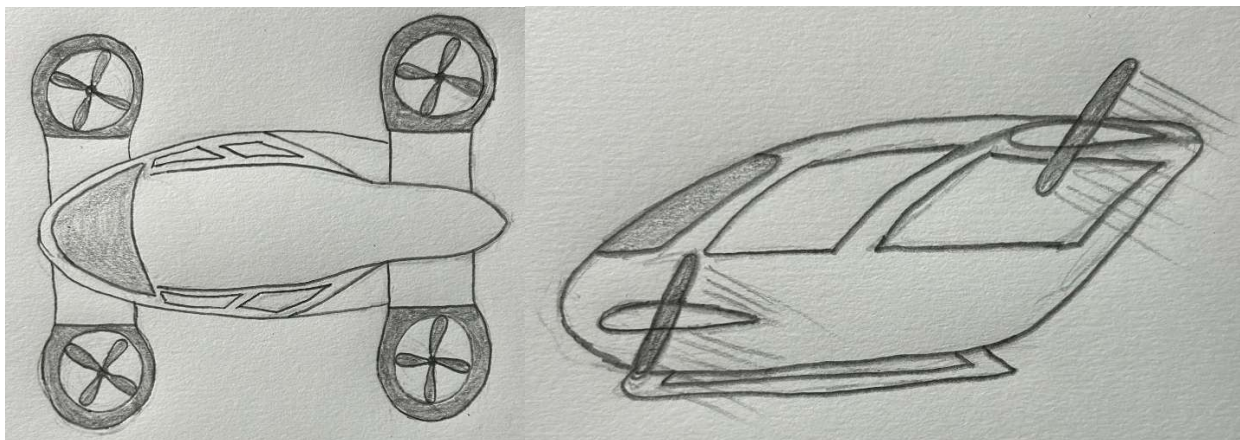


Figure 7 - Tilt rotor quadcopter

This design features a hybrid propulsion system powering four tilt rotors. This allows the use of a small, fixed wing in forward flight, creating a more efficient flight. A vertical offset between the forward and rear wing was added to decrease flow interference. A large, unblocked rear door was added to allow for easy unloading and loading of patients on rough and uneven ground. The total capacity of the aircraft is 3 personnel, one pilot, one medical personnel, and a patient.

2.1.2 Design 2 - Fixed Rotor Quadcopter

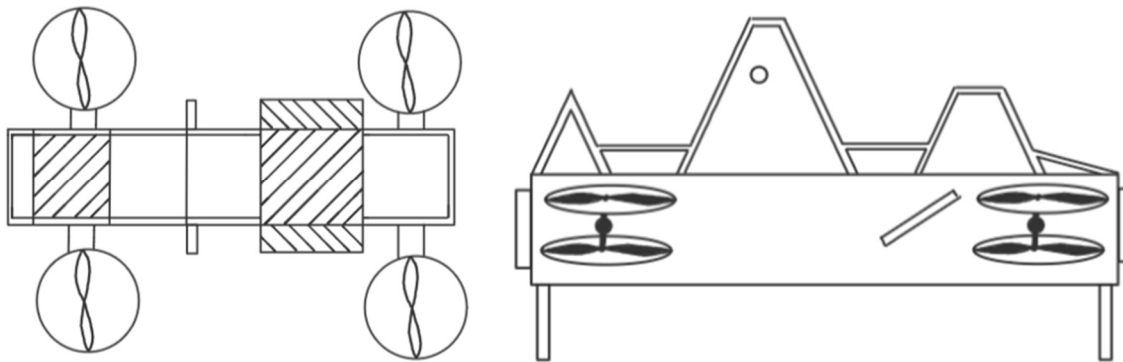


Figure 8 - Fixed rotor quadcopter

The Fixed Rotor Quadcopter is the smallest of the proposed designs allowing for access to hard-to-reach places. It has space for one passenger in the lower portion and one medical personnel sitting upright on top. Each arm has two motors/props in the event of a motor failure the aircraft will remain airworthy. The empty space in the front will contain a first aid kit or any other specialty equipment required. The very front and back contain a defibrillator and AED respectively. The main point of this design is maximizing maneuverability and reducing size while sacrificing capacity and range.

2.1.3 Design 3 - Lift and Cruise with Ducted Fans

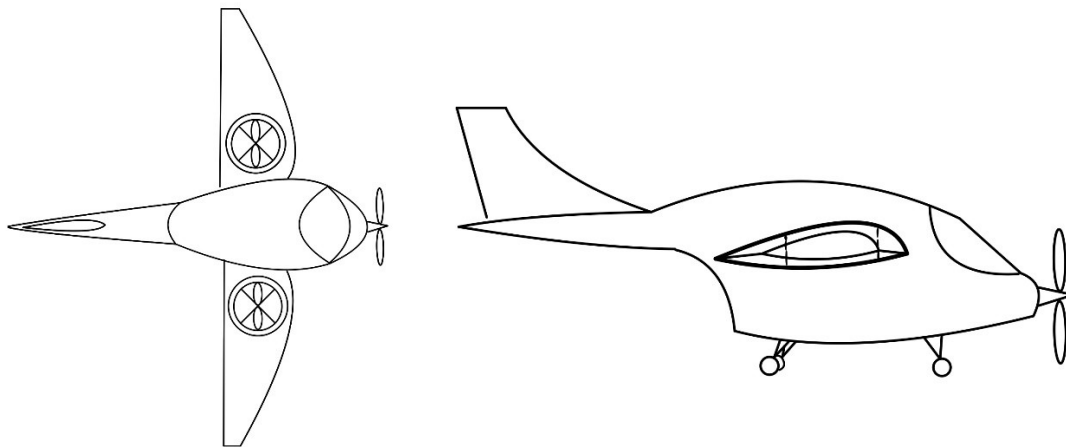


Figure 9 - Lift and cruise with ducted fans

The third concept is built off common lift and cruise aircraft designs. This concept features electric ducted fans located in each wing to enable vertical takeoff and landing, with an articulated design for precise control. Ducted fans were selected to create a safer environment

when landing or hovering at accident scenes. A forward propeller is located on the nose of the aircraft that will be engaged for forward flight. A turboshaft will be used to power the forward flight. Note that the forward flight propeller and ducted fans are supplied by two isolated power systems, allowing each to be optimized for their dedicated function.

2.1.4 Design 4 – Gyroplane

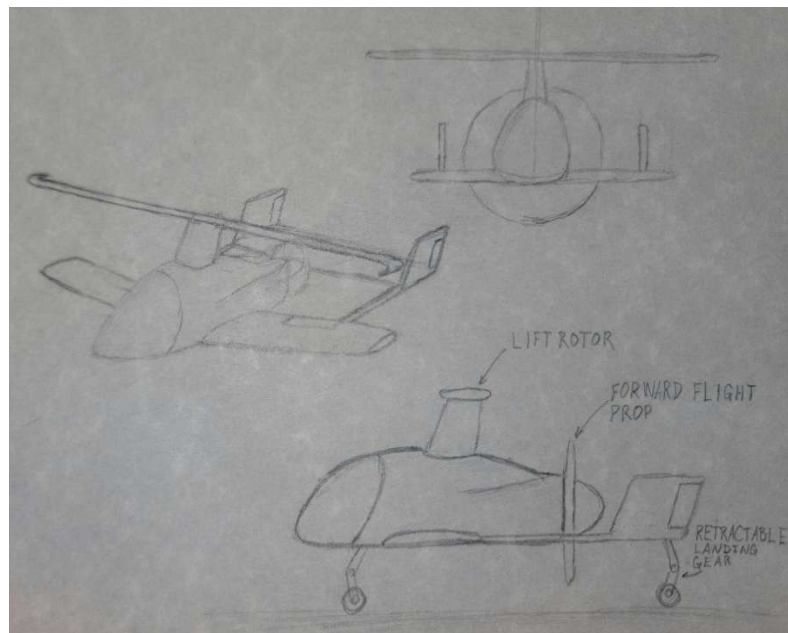


Figure 10 – Gyroplane

The gyroplane is a unique hybrid aircraft that combines powered and unpowered flight [5]. It uses a powered rotor for vertical takeoff, then transitions to unpowered autorotation during flight, making it efficient and stable. A rear pusher propeller provides forward thrust during cruise, allowing for smooth and controlled movement. Designed with a hybrid energy system, it can accommodate 2 to 4 passengers, making it a practical option for small transport missions. With fewer moving parts than most other systems, maintenance is reduced, increasing reliability and cost-effectiveness. One of its key advantages is safety—if power is lost, autorotation allows for a controlled descent, reducing the risk of uncontrolled crashes. Additionally, the front of the cabin opens, making it ideal for emergency scenarios where quick access to injured passengers is needed. Overall, the gyroplane offers a balance of efficiency, safety, and practicality, making it a strong choice for our application.

2.1.5 Design 5 - Tilt Rotor Micro Jet

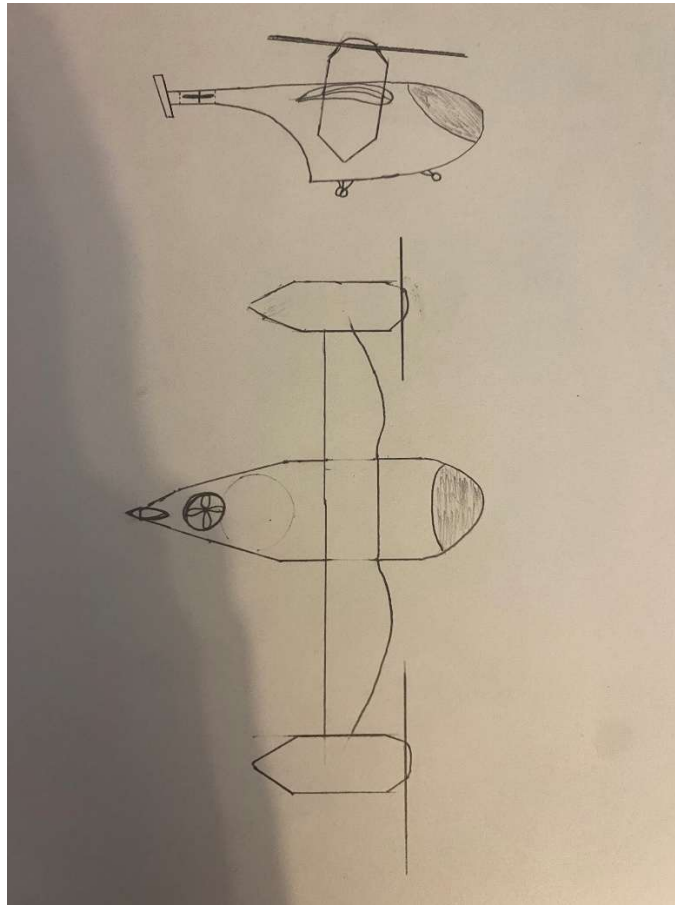


Figure 11 - Tilt rotor micro-jet

This tilt-rotor design takes inspiration from the V-22 Osprey. The concept features short, fixed wings with wingtip mounted, rotating propellers. Additional fan(s) can be introduced in the fuselage and wings to improve VTOL performance and stability. The aircraft would be suitable for a two-person medical team and would be capable of taking on 3-4 additional patients. The fixed wing is top mounted to improve ground visibility and mitigate ground impacts if the aircraft needs to land or hover around difficult terrain. The fixed wing improves the cruise speed and efficiency, both key performance specifications to address our need for good response time (high speed) and enough range to reach patients and return to the hospital base.

2.1.6 Design 6 – Dual Prop Lift and Cruise

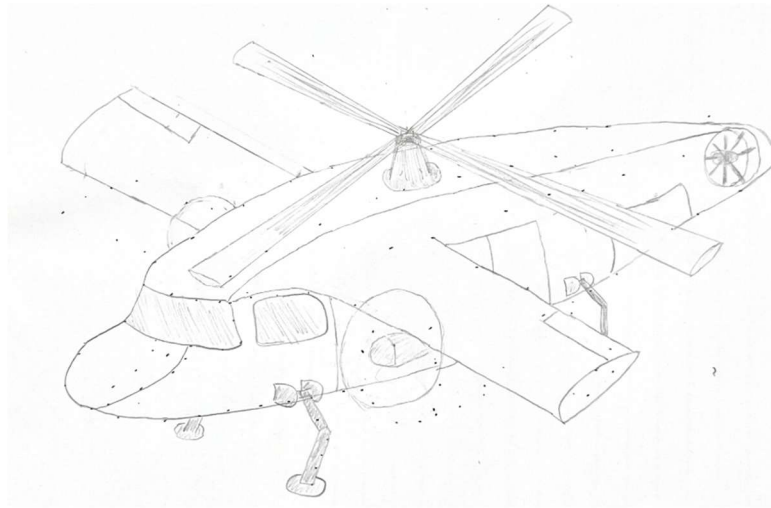


Figure 12 - Dual prop lift and cruise

The sixth design features a dual propeller lift and cruise design. It has a large top rotor that allows for helicopter like vertical flight. A fixed wing featuring two forward facing propellers allows for a transition to efficient forward flight after the HTOL has occurred. A large sliding door was added to allow for easy patient transfer as well as a integrated winch system for aerial rescue while hovering. Adaptive landing gear was a unique design concept that allows for the aircraft to land flat on non-flat surfaces, allowing more complex and difficult rescues.

2.2 Concept Selection

The design selection was initiated by selecting seven criteria that we believed was the most important for the design and success of our aircraft. This led us to choose speed, range, pollution, carrying capacity, total area, noise and cost. We ranked these versus each other, determining which of the criteria is most important to the final design. These rankings can be seen in Table 2.

Each of the criteria were ranked for each design, creating a matrix for each of the 7 categories. These can be seen in Appendix A. These values were imported into the aircraft design tool to select a final design, as seen in Figure 13. Figure 14 show the AHP output of the Aircraft Design Tool, which indicates the lift-cruise with ducted fans concept design is the most viable option, thus it was decided to proceed with further development of it.

Table 2 - Criteria ranking

Criteria	Speed	Range	Pollution	Capacity	Total Area	Noise	Cost
Speed	1	7/1	7/1	3/1	3/1	9/1	8/1
Range	1/7	1	3/1	3/1	3/1	9/1	5/1
Pollution	1/7	1/3	1	1/6	1/8	1	1/6
Capacity	1/3	1/3	6/1	1	1	7/1	1/2
Total Area	1/3	1/3	8/1	1	1	7/1	2/1
Noise	1/9	1/9	1	1/7	1/7	1	1/6
Cost	1/8	1/5	6/1	2/1	1/2	6/1	1

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    ],
    "categories": [

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Figure 13 - Weighted matrix ADT input

```

Command Window

Aircraft Design Tool Copyright (C) 2022 Mario Bras
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it under certain conditions;
see details in the included LICENSE file.

Concepts
- Lift-Cruise with Ducted Fans: 0.225571
- Tilt-rotor Micro Jet: 0.200586
- Gyroplane: 0.181598
- Dual Prop Lift-Cruise: 0.162350
- Fixed Rotor Quadcopter: 0.126676
- Tilt-Rotor Quadcopter: 0.103219

```

Figure 14 - Weighted matrix ADT output

The design was modified after initial selection of the concept to meet redundancy requirements, resulting in the refined concept shown in Figure 15. To enable pitch control, a tandem wing was added. With the original design of only two ducted fans, in the situation of a motor failure, there would be no redundancy while in VTOL operation. For this reason, additional ducted fans were added to both the main and tandem wing. Additionally, increasing the number of ducted fans used allowed for rotor area to be decreased, making it easier to locate them within the wings. This also allows for control surfaces behind the rotors on the trailing edge.

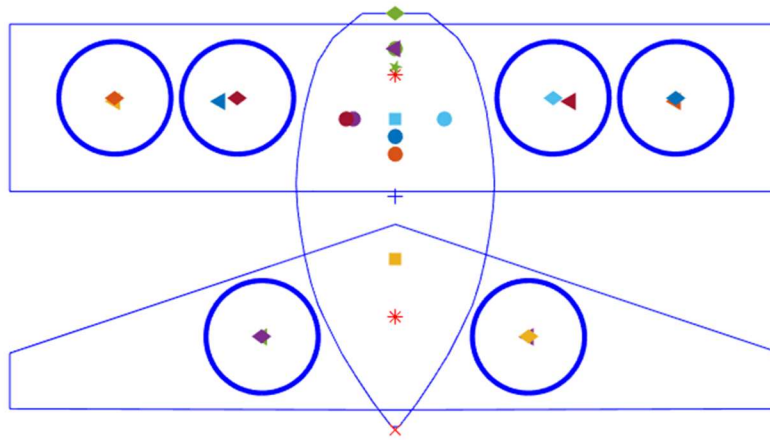


Figure 15 – ADT output of refined concept

3 Initial Sizing

With the preliminary design selected and refined, the next step was to determine the Maximum Take-Off Mass (MTOM) and determine the design points for vertical flight and forward flight. This process was highly iterative with the goal to maximize payload capabilities while minimizing power and empty mass of the aircraft.

3.1 Maximum Take-off Mass (MTOM)

The aircraft has a Maximum Take-Off Mass (MTOM) of 2453.61 kg, with weight distributed across four main categories. The empty mass makes up 54% of the MTOM, totaling 1284.10 kg, and includes all structural components such as the aluminum fuselage and wings, as well as the rotors and ducted fans. The battery accounts for 24% of the total weight, or 618.01 kg, and powers the six ducted fan motors during all vertical and forward flight phases except cruise. Payload is 500 kg, which includes four passengers at 100 kg each and 100 kg of medical equipment—consistent with the aircraft’s emergency medical transport role. Finally, 2% of the weight, or 51.50 kg, is allocated to jet fuel, which powers the turboshaft engine used only during

cruise. This distribution reflects a balanced compromise between energy efficiency, mission endurance, and safe operation across all phases of flight.

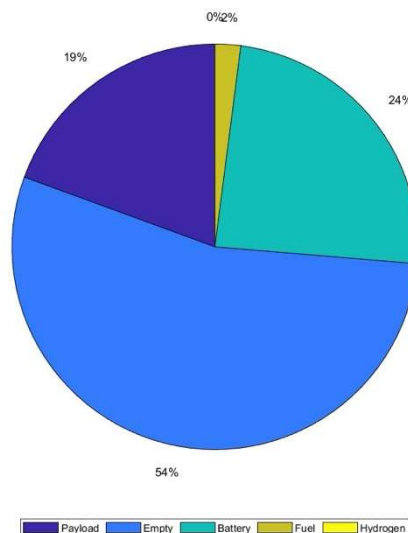


Figure 16 - Mass distribution chart

This breakdown represents an initial mass estimation using the design tool and does not account for secondary systems such as landing gear, avionics housings, or structural reinforcements, which could be refined in later design stages.

3.2 Design Points for Fixed-Wing (FW) and VTOL

The goal of this design point was to maximize the weight-to-power ratio (W/P) to reduce overall power consumption, while remaining within the constraints of the design space. This required carefully balancing the weight-to-disk area ratio (W/A) and wing loading (W/S) to ensure the rotors could support the vertical take-off and landing without requiring oversized wings.

Key considerations included fitting the rotors within the wing structure, satisfying stability criteria, and ensuring redundancy in case of a rotor failure. These constraints limited how far individual parameters like W/A, W/S, or W/P could be pushed. Rather than optimizing a single metric, compromises were made to ensure stable and efficient flight throughout the entire mission.

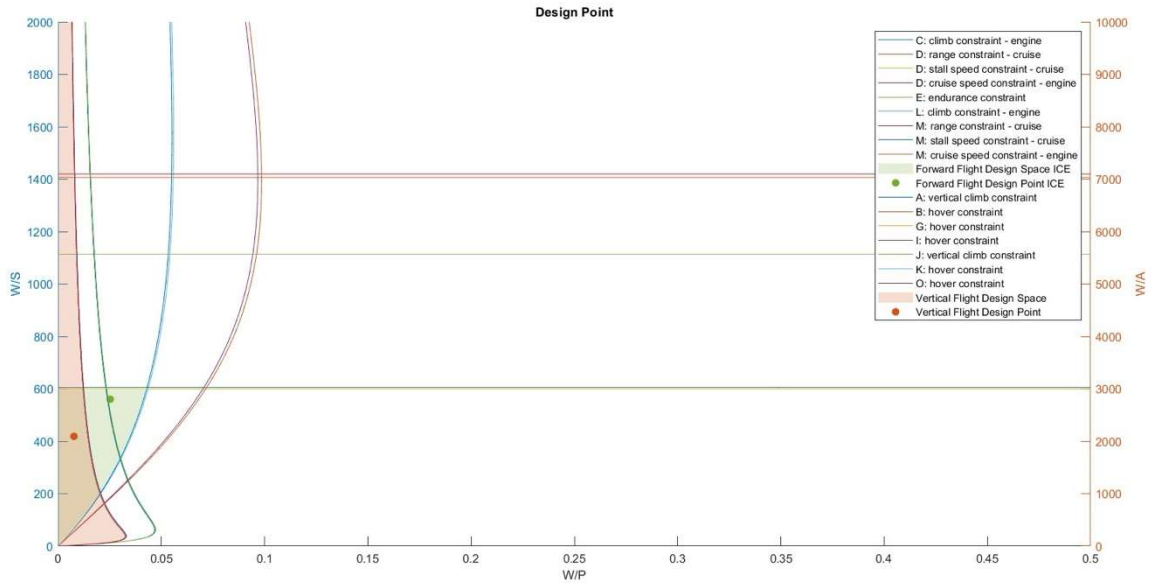


Figure 17 - Aircraft design space with final design points

The final design has a weight-to-power ratio of 0.0076 N/W and a weight-to-disk area ratio of 2093.98 N/m² during vertical flight. For cruise, the weight-to-power ratio is 0.0252 N/W, and the wing loading is 560.52 N/m². These values represent a well-rounded solution that meets performance targets while minimizing energy demands and maintaining operational safety.

4 Wing Design

4.1 Airfoil Selection

Due to the added drag and non-uniform surface introduced by the embedded ducted fans, minimizing profile drag during forward flight was a key concern in airfoil selection. The NACA 6-series was chosen over other families for its ability to maintain extended laminar flow, resulting in lower drag at moderate Reynolds number [6]. The NACA 63-210 was ultimately selected for its low drag, favorable lift characteristics, and consistent 10% thickness, which enables simple integration of the fans along the wingspan while helping to reduce flow disruption and maintain aerodynamic efficiency [7].

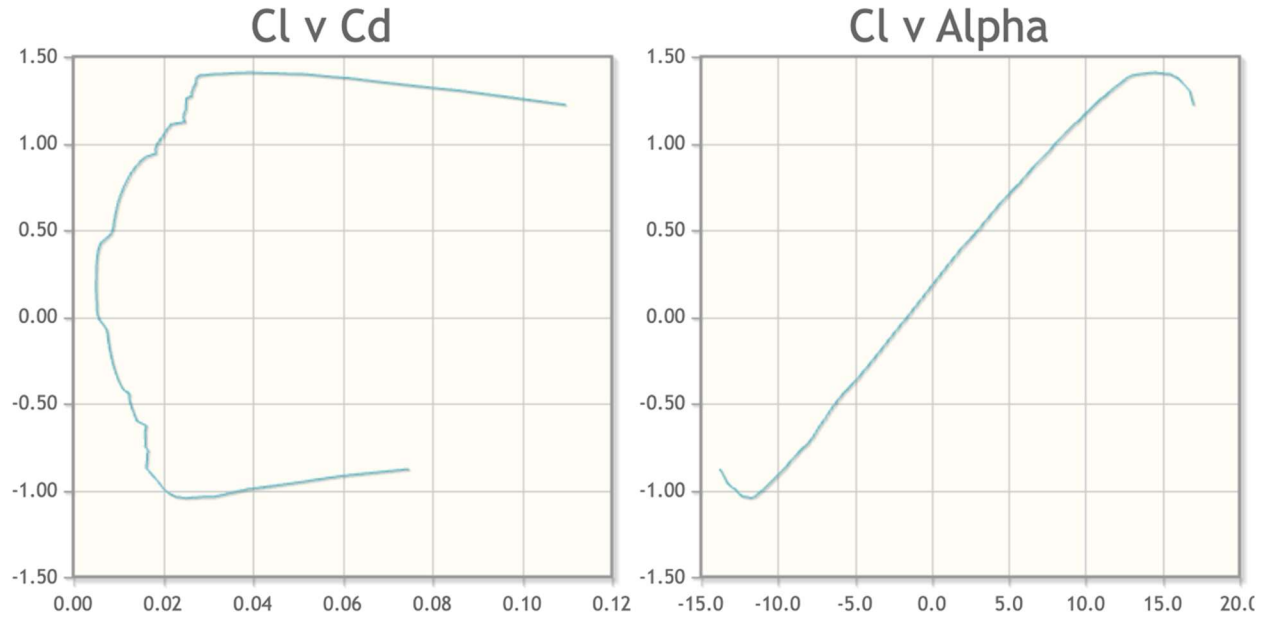


Figure 18 - NACA 63-210 airfoil performance graphs

This choice also aligned with reference aircraft such as the Viking Dragonfly and Quickie Q2, which feature tandem wing configurations and similar geometric constraints. The airfoil provides predictable behaviour near stall, a smooth lift curve slope, and maintains efficiency in the cruise region.

4.2 3D Wing Design

The 3D design of the main and tandem wings was developed to balance aerodynamic performance, packaging constraints, and longitudinal stability requirements. While the 2D airfoil selection focused on characteristics like thickness and drag, the 3D geometry had to account for integration with the fuselage and embedded ducted fans.

One of the primary constraints was maintaining a sufficient chord length to fully enclose the ducted fans, without increasing the fuselage length. This led to fixed span and chord dimensions for both wings, with the main wing having a span of 11 m, a taper ratio and sweep angle of 0, while the tandem wing also spans 11m but includes a taper ratio of 0.3 and a sweep angle of 18.43°.

The tandem wing design includes both taper and sweep, features introduced to improve the aircraft's longitudinal stability. The tandem wing was modeled as a horizontal tail in stability calculations, where introducing taper ratio increased the m parameter and decreased the r parameter from course graphs. This effectively reduced the downwash gradient $\frac{de}{d\alpha}$, helping the aircraft achieve a stable configuration. Through an iterative process, the tandem wing was finalized with an aspect ratio of 6.41, while the main wing was designed with an aspect ratio of 4.62.

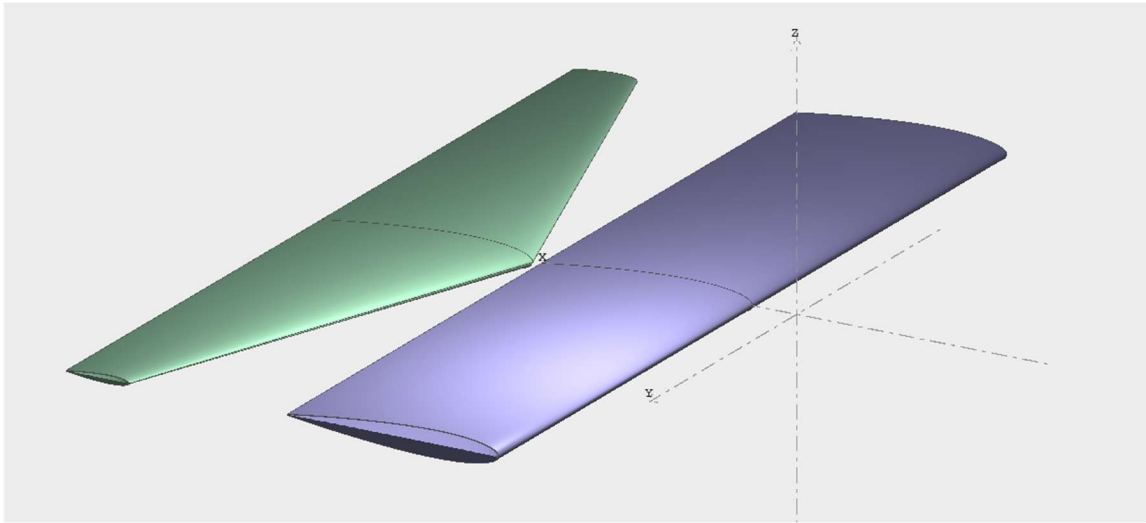


Figure 19 - 3D models of main and tandem wing

4.2.1 Ducted Fans Covering

To decrease drag over the ducted fans and increase lift during forward flight, a variety of fan covers were explored. Shown in Figure 20 are four cover concepts that were explored, taken from a similar existing application. These included an iris-shutter, sliding cover panels, fan shutters, and large single doors.

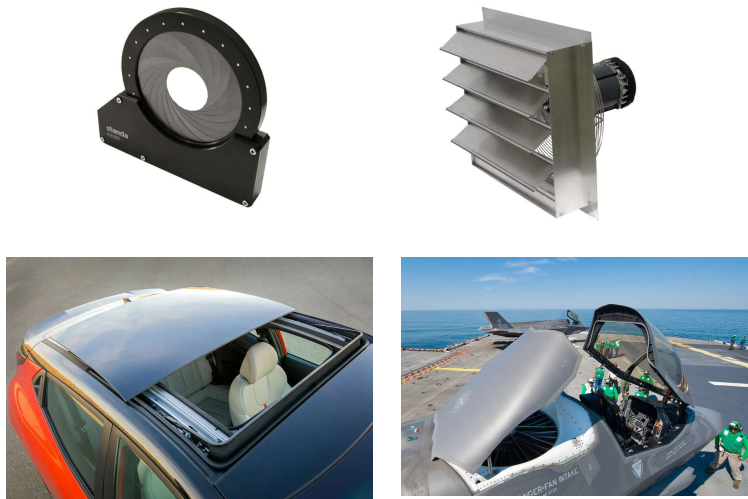


Figure 20 - Ducted fan covering concepts [8] [9] [10] [11]

The shutter-style covers were selected for their simple integration and minimal impact on existing components. This design allows the surface of the wing to be almost completely unchanged when the shutters are closed, creating an efficient wing. When the shutters are open, it will allow the rotors within to produce the needed lift. Other explored options were found to

have challenges with integration of the cover in both the main and tandem wing while also adding excessive amounts of air resistance when transitioning to forward flight. The final integrated shutter design can be seen in Figure 21 in the vertical flight configuration, and in Figure 22 in the forward flight configuration.

It is understood that this solution, while the most ideal of all those explored, may have issue with added air resistance during this transition, but we believe that it may be functional if properly designed. The flaps close in the direction that airflow would naturally put them, allowing the transition to be more seamless and the mechanism to be more robust. Within the tandem wing, the curves of the tapered wing are more complex, creating more complex panels for the covers and thus requiring more meticulous attention during further development. It should also be noted that while the shutters improve lift and reduce drag, imperfections in joints at the shutter will result in a less efficient wing in the closed configuration. Additionally, to accommodate the fans within the wings, a lightweight metal frame would run inside the wing structure to support the fans.

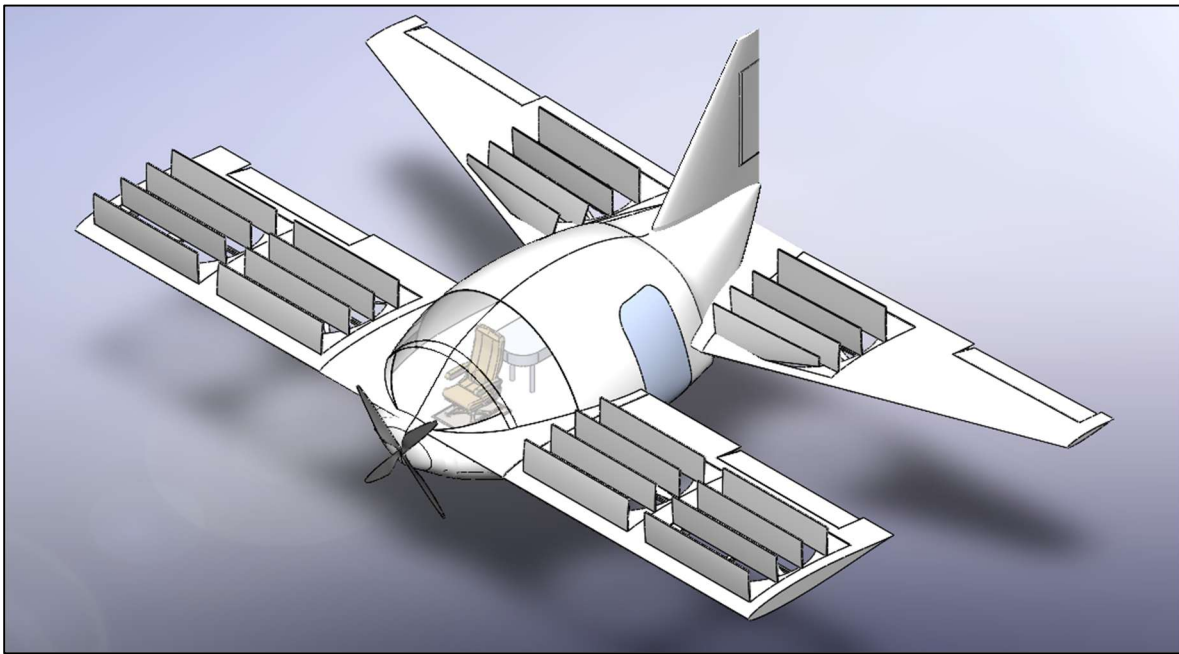


Figure 21 - Ducted fan covers open

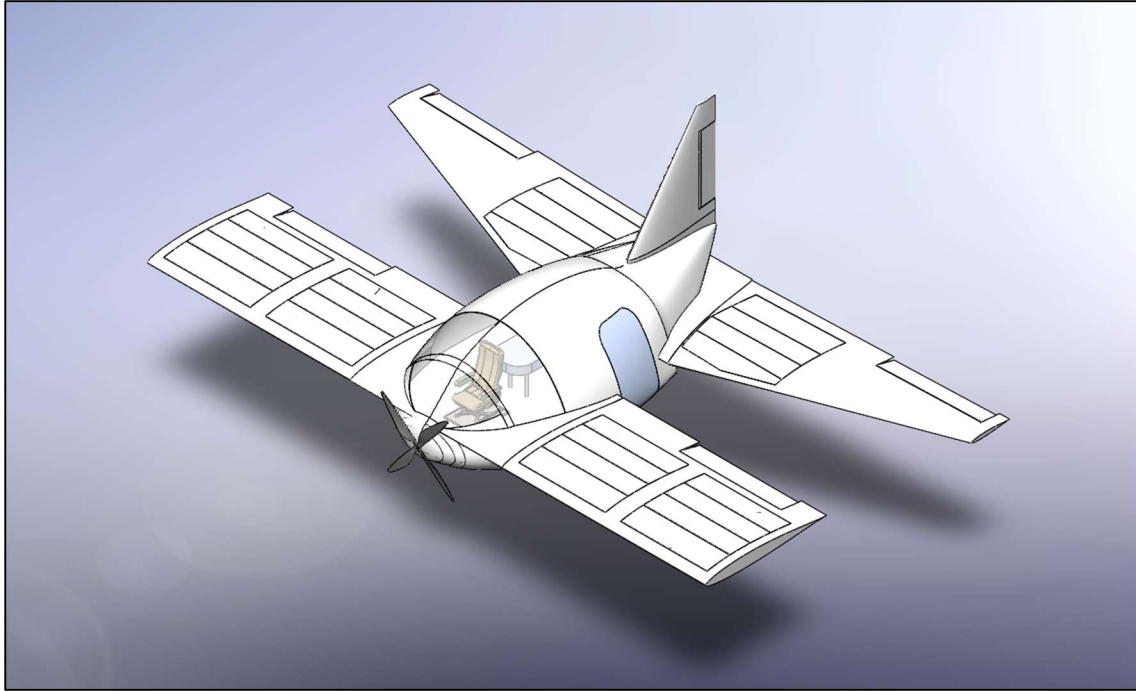


Figure 22 - Ducted fan covers closed

4.3 Lift and Drag Estimation

An aerodynamic analysis was done using XFLR5, a software tool for airfoil and wing performance evaluation, to assess the lift, drag, and stall characteristics of the tandem wing configuration.

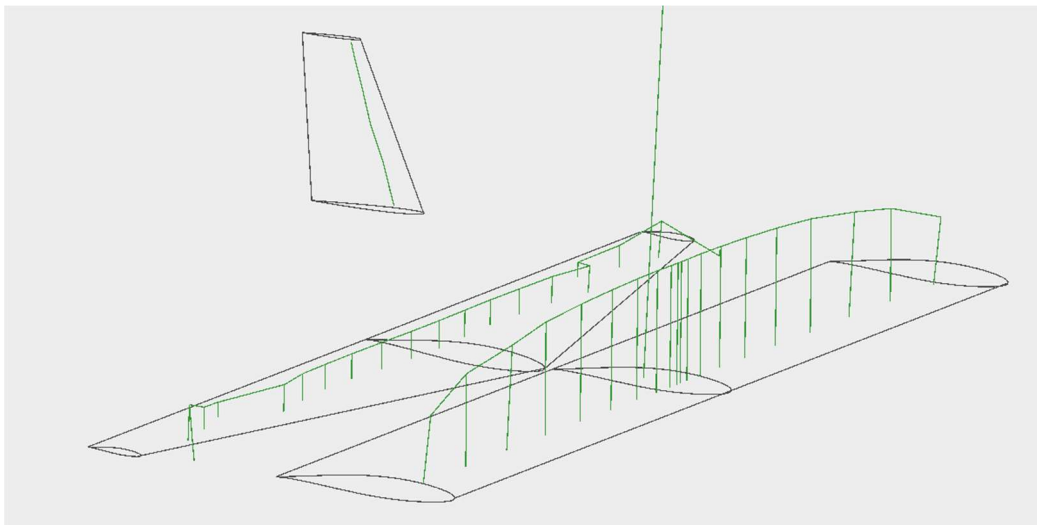


Figure 23 - Lift Distribution from XFLR5 Analysis

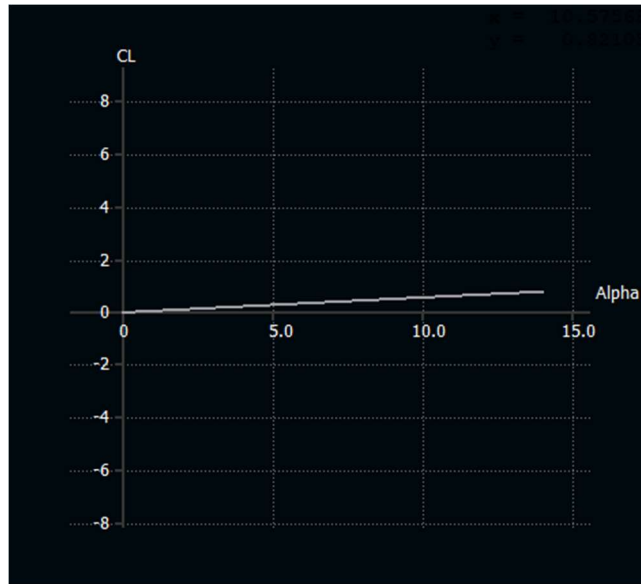


Figure 24 - C_L vs α Polar Plot from XFLR5 Analysis

The front wing stalls at approximately 14.5° , while the rear wing continues to generate lift up to 18.5° . This behavior is desirable in tandem wing configurations, since having the front wing stall first helps maintain pitch stability and prevents sudden loss of control. When the front wing loses lift before the rear, it naturally causes the nose to drop, reducing α and allowing recovery from stall. If the rear wing stalled first, the aircraft could become unstable, with the nose pitching up further and deepening the stall condition. The rear wing's stall is not shown in the graph because the analysis ends once the maximum lift coefficient of the main wing is reached.

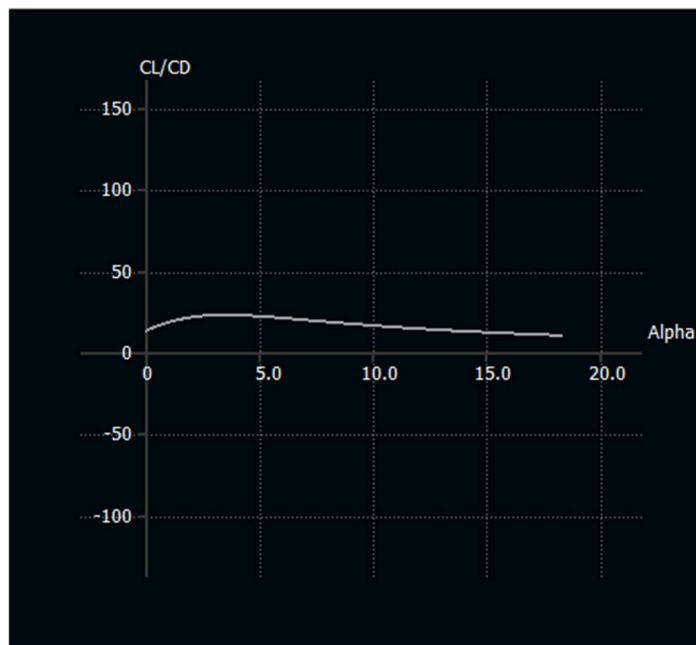


Figure 25 - C_L/C_D vs α from XFLR5 Analysis

The lift-to-drag ratio of 20.5 at level flight is similar to the reference aircrafts, the Viking Dragonfly and Quickie Q2, indicating strong aerodynamic efficiency. Since the ducted fans were not included in this analysis, the actual drag during cruise would be slightly higher, but the result still demonstrates effective cruise performance.

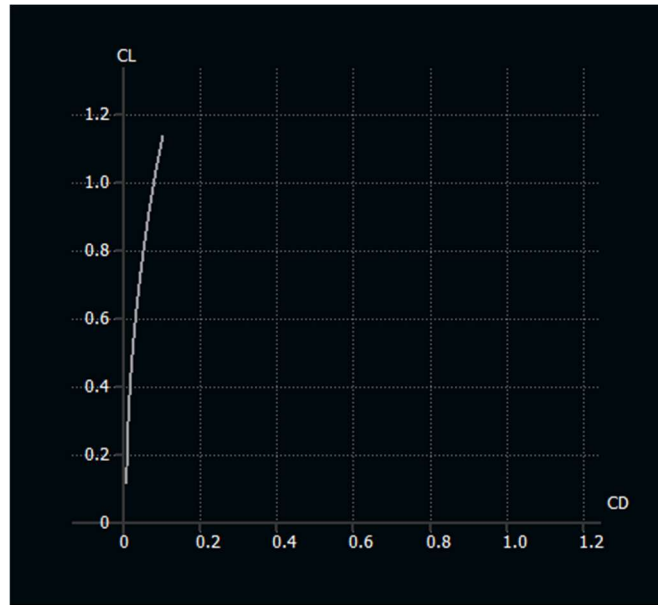


Figure 26 - C_L vs C_D from XFLR5 Analysis

This C_L vs C_D curve shows a steady lift increase with minimal drag rise. The absence of sharp changes suggests the wing operates well within safe limits, offering predictable behavior during cruise. However, since ducted fans weren't included in this analysis, real-world drag would be higher, slightly reducing efficiency.

4.4 Shear/Bending Moment Diagrams

The bending moment and shear force diagram was created for each of the two wings, the main wing, and the tandem wing. Since our aircraft undergoes separate stress profiles during VTOL and cruise, a separate diagram has been created for each of these situations. All the forces were calculated based on the masses estimated during research as well as calculations for total mass and lift.

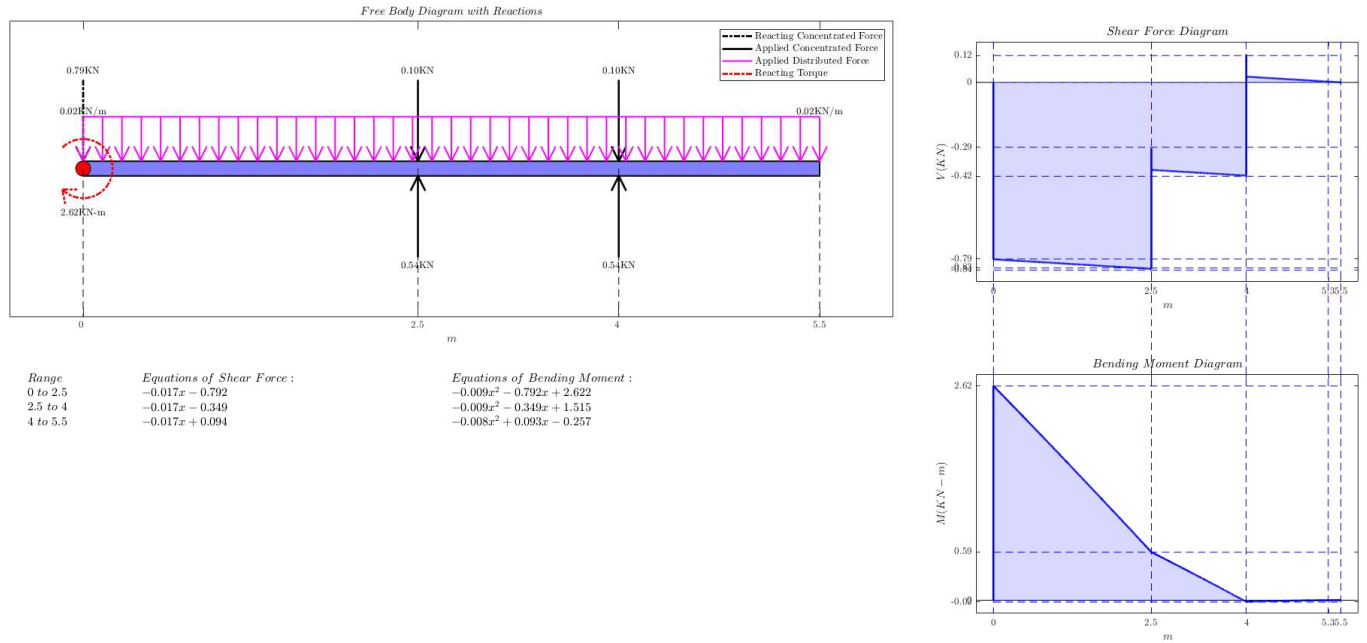


Figure 27 – Main wing diagram in VTOL

For VTOL, the lifting force comes from the ducted fans, placed at the two locations on the wing. The distributed mass of the wing, and the mass of the motors was added as well.

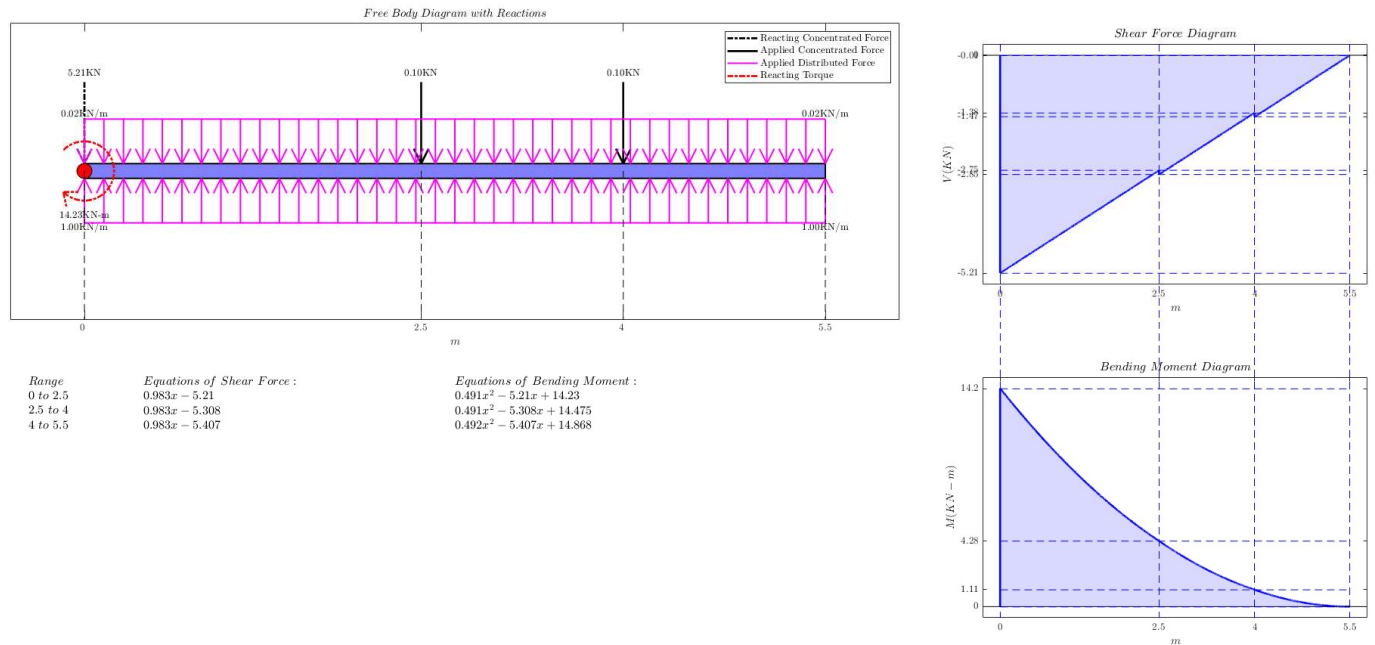


Figure 28 – Main wing diagram in cruise

In cruise, the lifting force was assumed as a uniform distributed load spanning the wing. The same weight of the wings and motors was used in cruise.

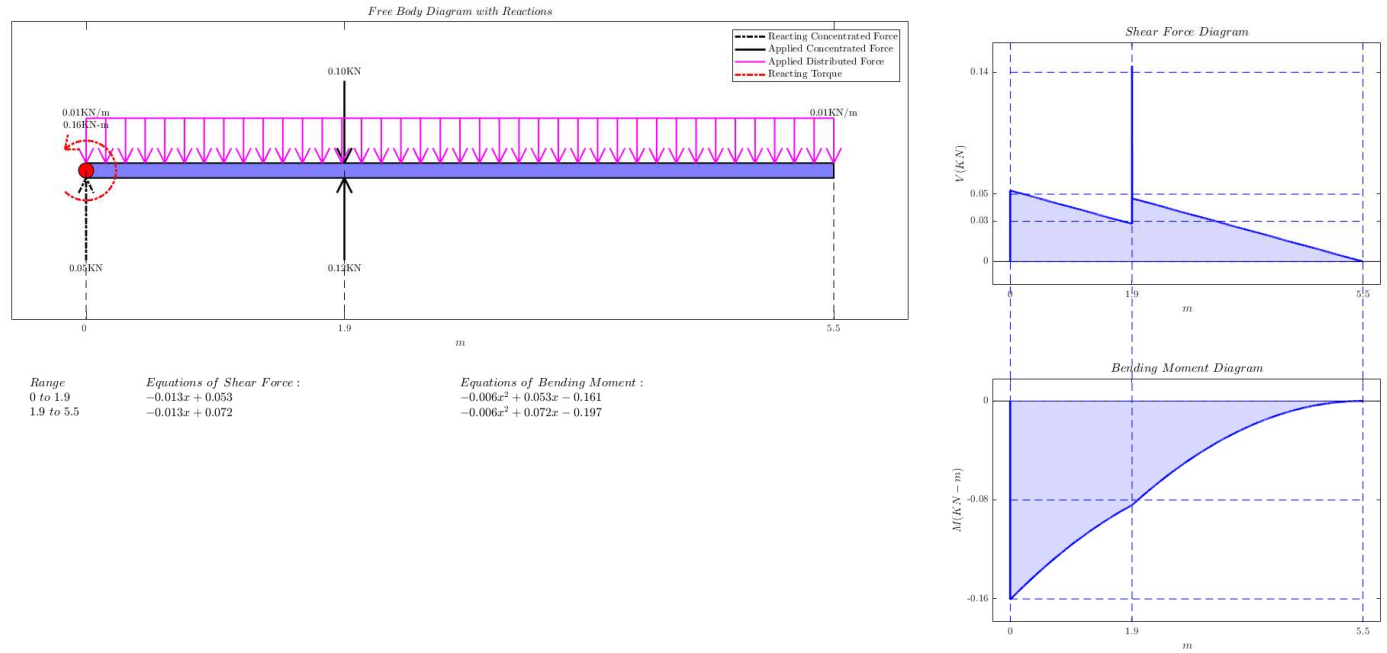


Figure 29 – Tandem wing diagram in VTOL

The Tandem wing features one less rotor, reducing the mass as well as the number of vertical forces applied through the wing in VTOL. The Tandem wing in cruise has a very similar setup to the main wing, with the exception of one less load from the rotor assemblies.

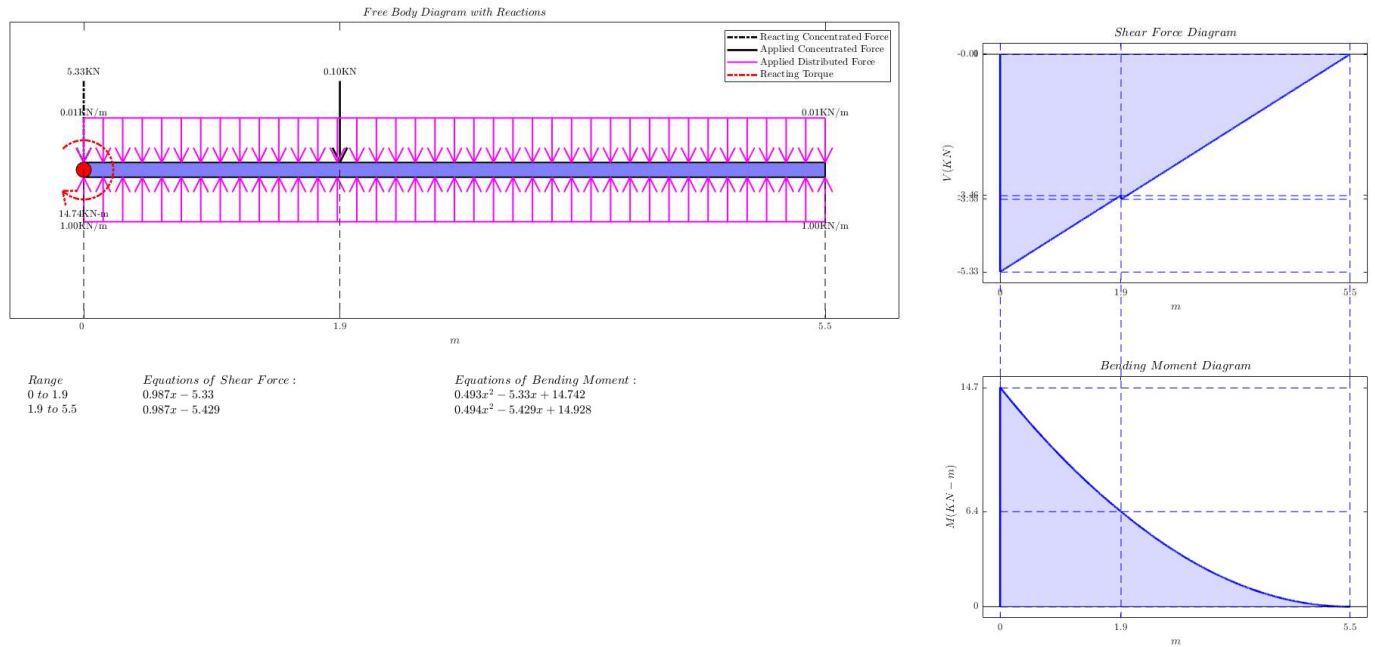


Figure 30 – Tandem wing diagram in cruise

5 Propulsion

5.1 Powerplant Layout/Description

The vertical flight propulsion system for the aircraft is the electric ducted fans located on the main and tandem wings. The horizontal propulsion system is a turboprop engine and propeller placed at the front of the aircraft. The vertical flight (take-off & landing) and horizontal (cruise) flight conditions and details are summarized below.

The following table details the vertical takeoff and landing flight plan for energy considerations. The maximum rate of climb and time for each section of flight. The aircraft will vertically climb then transition into cruise. Once descended after cruise the aircraft will transition back into vertical landing, a hover segment is included. The total flight time is for a complete round trip, the required time is the minimum time needed, and the design time is for the energy estimation.

Table 3 – Vertical Flight Plan Breakdown

Total Transition Time [s]	60
Total Landing Time [s]	150
Total Hover Time [s]	160
Total Required Time [s]	520
Design Time [s]	1040
Design Time [min]	17.333
Design Time [hours]	0.2889

Note that the design time is double the required flight time as, this is to account for efficiency & degradation losses in the battery. As the aircraft will be operating in temperatures below -40°C the loss of energy effects batteries exponentially more than at room temperature. The time to climb and descend are also assumed to take an equal amount of time and energy even though the decent will take less energy. The transition time is also very conservative for the energy calculations; the electric motors will likely not be operating at maximum power during the transition.

For the power plant layout, the batteries will be connected to inverters then to the electric motors and props while the fuel tank and turboprop engine is a separate system. See the diagram below.

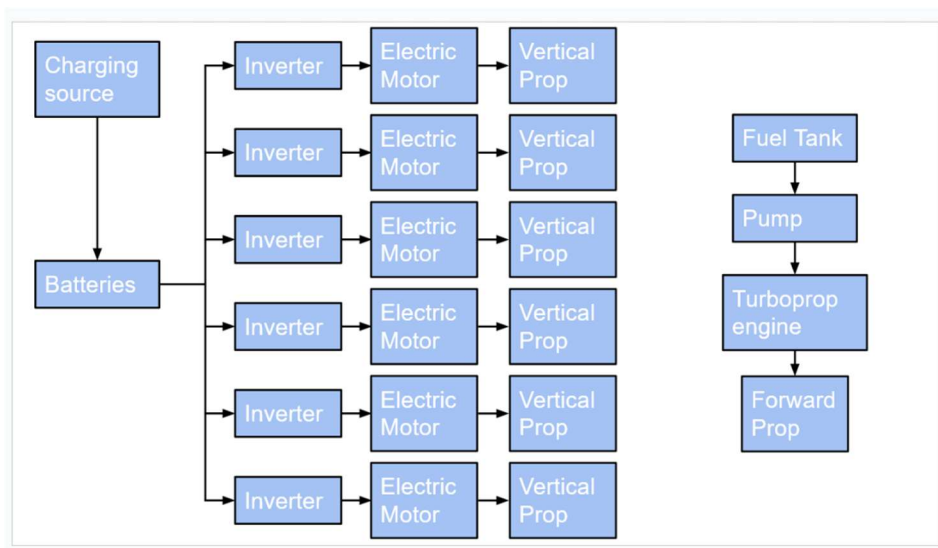


Figure 31 - Powerplant Layout

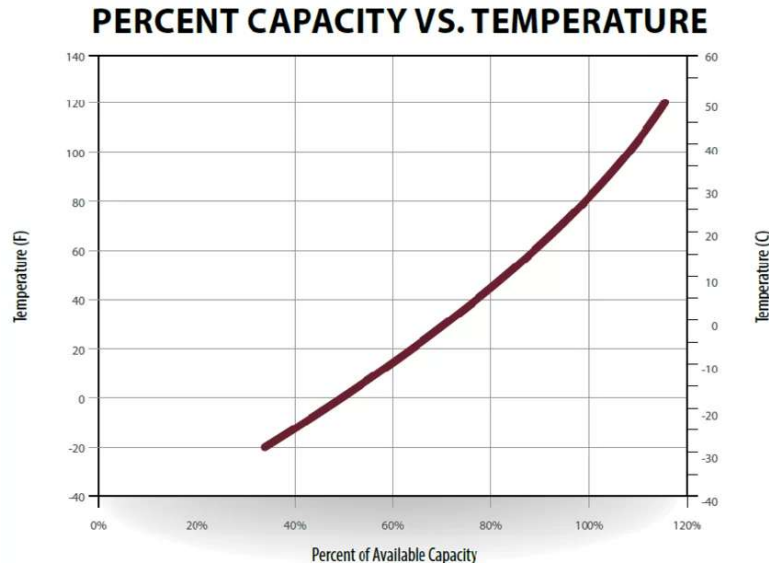


Figure 32 - Battery capacity vs temperature [12]

To help counteract some of this loss the battery will be insulated as well as kept warm by the excess heat exhausted by the turbo prop in cruise. During vertical take-off and landing the battery will keep itself sufficiently heated from the excess heat generate by its operation.

The following table details the cruise flight plan for fuel considerations. The maximum time, range and speed to the destination and return. The aircraft will be expected to climb slowly to cruise conditions after vertically climbing. It will remain in cruise conditions until decent where it will transition into vertical landing.

Table 4 – Cruise Flight Plan Breakdown

Range [km]	200
Speed [m/s]	60
Flight Time [hr]	0.93

5.2 Technical Specifications of Components

All propulsion systems were initially sized generally and iterated upon throughout the design process. The power, fuel and battery estimations presented below were calculated assuming a maximum take off mass of 2400 kg. Further details and calculations can be found in the calculation's spreadsheet in [13] and summarized as tables in Appendix B.

Table 5 – Electric Motor Specifications

Number of Motors	6
Minimum Power per Motor [kW]	167.1
Total Motor Power [kW]	1002.5
Sized Motor Power [kW]	340.0
Weight per Motor [kg]	34.0
Total Sized Motor Power [KW]	2040.0

The electric motors were sized with reference to Lucid Motors who have recently developed a new electric motor for Formula E capable of producing 355 kW in just 32 kg [14]. The total power sized for vertical climb is double that of the required for redundancy. One ducted fan would be able to fail on each of the front main wings. It would also be possible for one ducted fan to fail on the back tandem wing while still maintaining control and power to climb.

The rotors were sized during the design process based on similar industry standards and iterated accordingly to performance requirements. The efficiency and speed were maximized to minimize energy requirements, the diameter was a constraint to fit inside the wings.

Table 6 – Rotor Specifications

Diameter [m]	1.2
Efficiency	85.00%
Tip Speed [m/s]	250
Blades	5

For forward flight the turbo prop was sized to operate at roughly 60% max power to reduce strain on the engine during cruise and leave extra power if needed. A turbo prop was selected for its high power and small form factor.

Table 7 – Engine Specifications [15]

Required Engine Power [kW]	111.9
Engine	PBS TP100
Weight [kg]	61.6
Power [kW]	180
Height x Width [mm]	398 x 330
Length [mm]	891

The propeller was selected in parallel with the engine, like the rotors. Typical values were chosen and then iterated based on the performance and requirements for the cruise. Efficiency and speed were maximized and while the diameter was constrained by the aircrafts size.

Table 8 – Propeller Specifications

Diameter [m]	2
Efficiency	87%
Tip Speed [m/s]	197.8
Blades	4

As the aircraft uses both electric and chemical propulsion it must carry sufficient battery and fuel. For the fuel we selected Jet A-1 as it has a much lower freezing point, ideal for winter and northern conditions. It is also the jet fuel of choice in the rest of the world.

Table 9 – Fuel Estimate

SFC [kg/kwh]	0.5
Required Mass [kg]	51.9
Reserve + Trapped	20%
Total Mass [kg]	62.2

The specific fuel consumption is sourced from the engine manufacturer [15]. The amount of reserve fuel was chosen to be adequate for an emergency where the aircraft may have to deviate from its route or required additional time to find a suitable landing location.

For the battery energy estimation some assumptions were made for the vertical flight plan. All calculations were done assuming the energy required was for climb to simplify calculations. The total designed flight time was also doubled from the required flight time to account for energy losses in the system as discussed in the powerplant layout/description section 5.2.

Table 10 – Battery Estimate

Density [kWh/kg]	0.45
Total Energy [kWh]	289.6
Total Weight [kg]	643.6
Total Package [kg]	700
Density [kWh/L]	300-700
Volume[L]	530

The battery is lithium ion as its energy density and long lifetime are currently unmatched. As the weight of the battery is very significant to the aircraft, we assumed a very high energy density, while commercially available it is not as economical as less dense alternatives [16]. Further developments in battery technology are likely to improve the performance. Development for a specialized battery pack may be required. Note the weight of the total battery package includes a rough estimate for the casing and insulation.

6 Fuselage and Structural Design

6.1 Internal Layout

The internal layout of the fuselage allows for the pilot to sit at the front of the aircraft, two medical personnel to sit on either side of a stretcher containing the passenger. Estimates for seating size were made based on short distance passenger seating standards, and a reference

stretcher size was used to ensure adequate room to accommodate it [17]. Additionally, the reference area of a typical ground ambulance patient compartment was included [18].

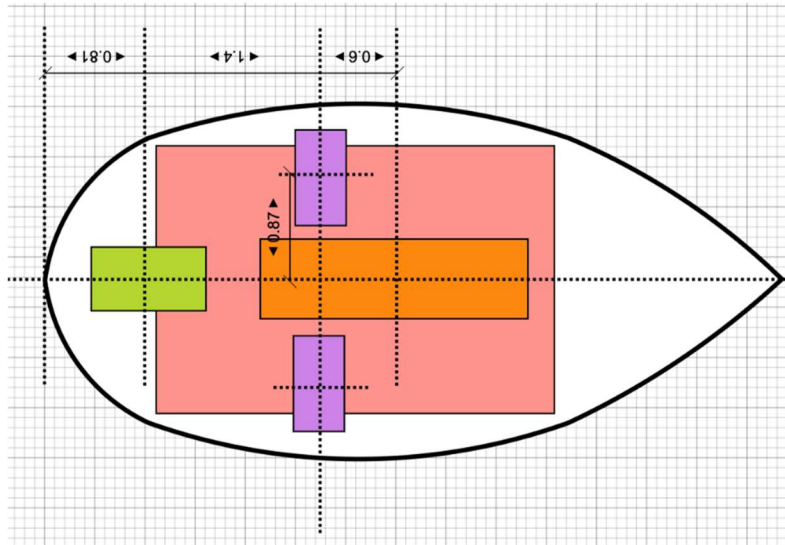


Figure 33 – Fuselage internal layout

Seen above, the pilot seating is shown in green, the passenger seating is shown in purple, and the stretcher is represented by the orange rectangle. The red rectangle represents the ideal ambulance bay area needed within the aircraft as well.

6.2 Fuselage Shape

The fuselage shape was chosen to adequately surround the required seating described in the previous section. This resulted in the shape seen below. We made the nose of the aircraft rounded to reduce drag. The widest part of the aircraft is 2.83m, around halfway between the ends, and the overall length of the fuselage is 5.88m.

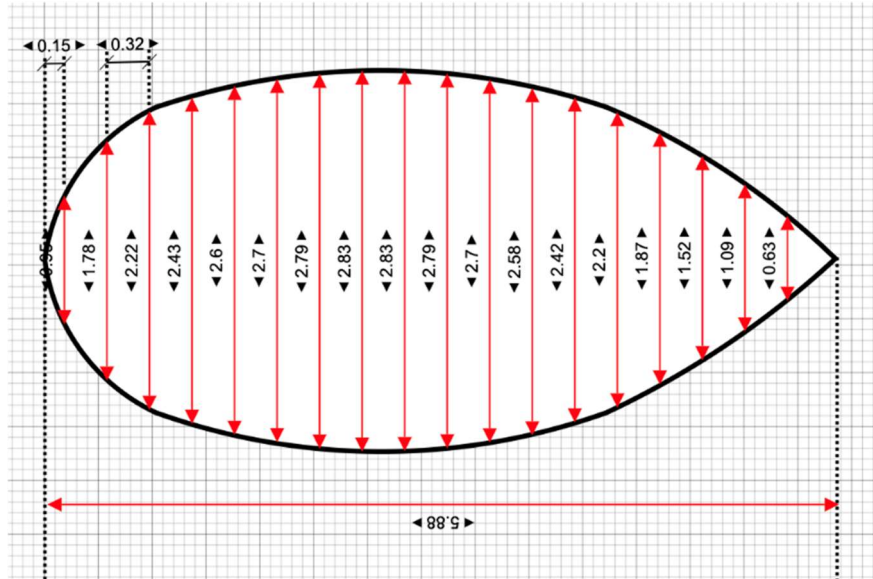


Figure 34 - Fuselage dimensions

Following the initial design of the fuselage, structural analysis was performed on it to provide insight to the loading it experiences. The fuselage bending and shear moment diagram can be seen below. The weights of each mass included in the ADT was added at its respective locations, as well as an estimate for the distributed mass of the fuselage itself. The reaction forces at each of the wings demonstrates how our center of gravity is forward of the geometric center as expected.

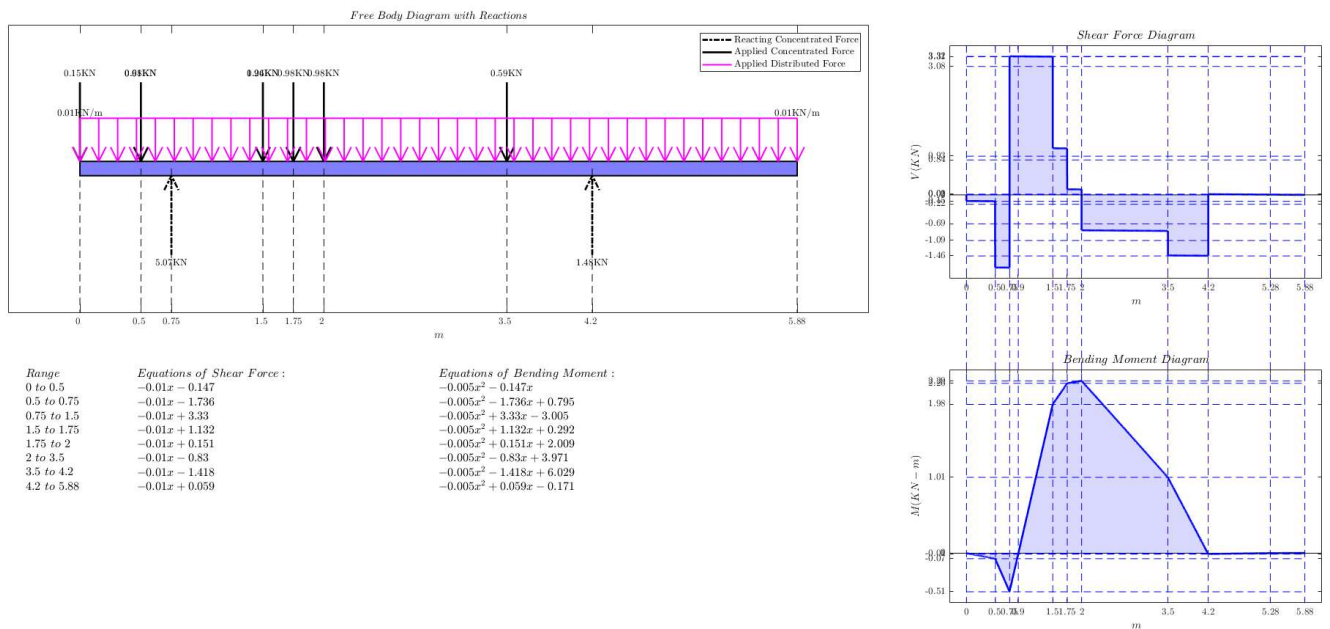


Figure 35 - Fuselage bending/shear diagram

6.3 Drag Estimation

The fuselage drag estimation included calculating the total frictional force along the wetted surface of the fuselage and converting it to a zero-lift drag coefficient. The drag estimation accounts for both the nose and main body sections of the aircraft.

The wetted area was determined using the projected top and side areas and applying the correction factor, while the friction coefficients were based on Reynolds number analysis. The surface roughness was factored in using correction formulas provided in the course notes, with metal shell assumed as the surface type. The Reynolds number for the main body indicated fully turbulent flow, while the nose was assumed to be laminar, with an increased coefficient to reflect curvature effects. A summary of the key inputs is provided in Table 11.

Table 11 – Key Parameters Used in Fuselage Drag Estimation

Parameter	Value	Source/Method
Cruise Velocity	60 m/s	Pre-determined
Wetted Area	24.28 m ²	Calculated from top and side projections
Surface Type	4.05E-06 m	Metal Shell from table in Lecture 10
Fuselage Length	5.74 m	CAD Geometry
Reynolds Number	1.15E+08	Calculated from flow conditions
Friction Coefficient	0.0083	Calculated
Form Factor	1.058	Calculated using course formula
Interference Factor	1.0	Assumed based on isolated body
Total Frictional Force	476.31 N	Calculated
Base Drag Coefficient	0.0120	Result

The resulting base drag coefficient for the fuselage is $C_{D_0} = 0.120$. This value is incorporated into the total drag buildup for the complete aircraft during performance and propulsion analysis.

7 Sizing of Vertical and Horizontal Tails

7.1 Airfoil Selection

The vertical tail airfoil selection focused on ensuring equal control during both positive and negative rudder deflection. Symmetrical airfoils were considered ideal for this application, with emphasis placed on low drag and structural ability. A thickness-to-chord ratio between 9-12% was identified as optimal, allowing for sufficient stiffness and reduced drag [19]. These criteria led to a range for NACA airfoils from 0006 – 0024. Among these options, the NACA 0012 was chosen due to its widespread use in similarly sized general aviation aircraft such as the Cessna 152, 172 and 182, and its favorable aerodynamic profile, offering consistent performance near stall.

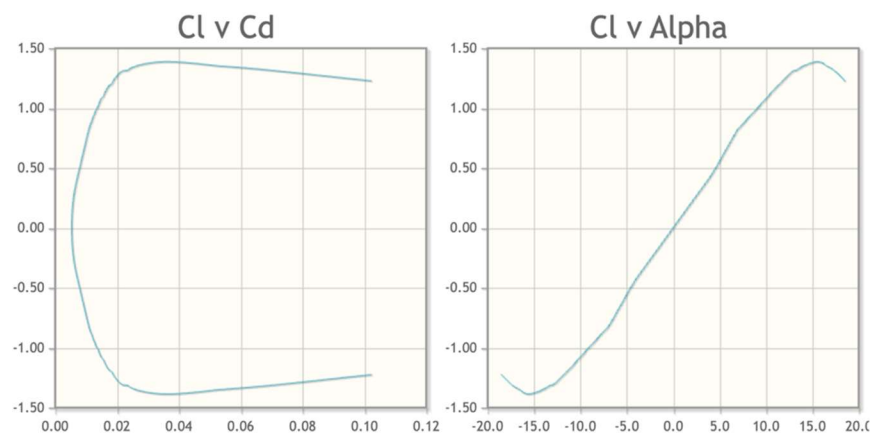


Figure 36 - NACA 0012 Airfoil Performance Graphs

7.2 3D Design

While the 2D design of the vertical tail, discussed in Section 7.1, focused on selecting a symmetric airfoil with appropriate thickness-to-chord ratio and camber characteristics, the 3D design process addressed the overall shape and geometry of the tail to meet stability targets and aerodynamic efficiency.

The initial design used a leading-edge sweep angle of 20° , a value commonly applied to aircraft of similar size [20]. The initial design also used an aspect ratio of 1.3 and taper ratio 0.4, which were typical values for vertical tails [21]. From there, key parameters such as sweep angle, aspect, height, and taper ratio were refined through an iterative process aimed at achieving a directional stability coefficient within the recommended range.

After multiple design loops, the final vertical geometry included a leading-edge sweep angle of 35° , an aspect ratio of 1.5, a taper ratio of 0.5, and a total tail height of 1.95 meters. These values provided the required directional stability while maintaining low drag and a

compact form factor. Since there were no significant or spatial constraints on the tail design, the focus remained on aerodynamic effectiveness and overall stability.

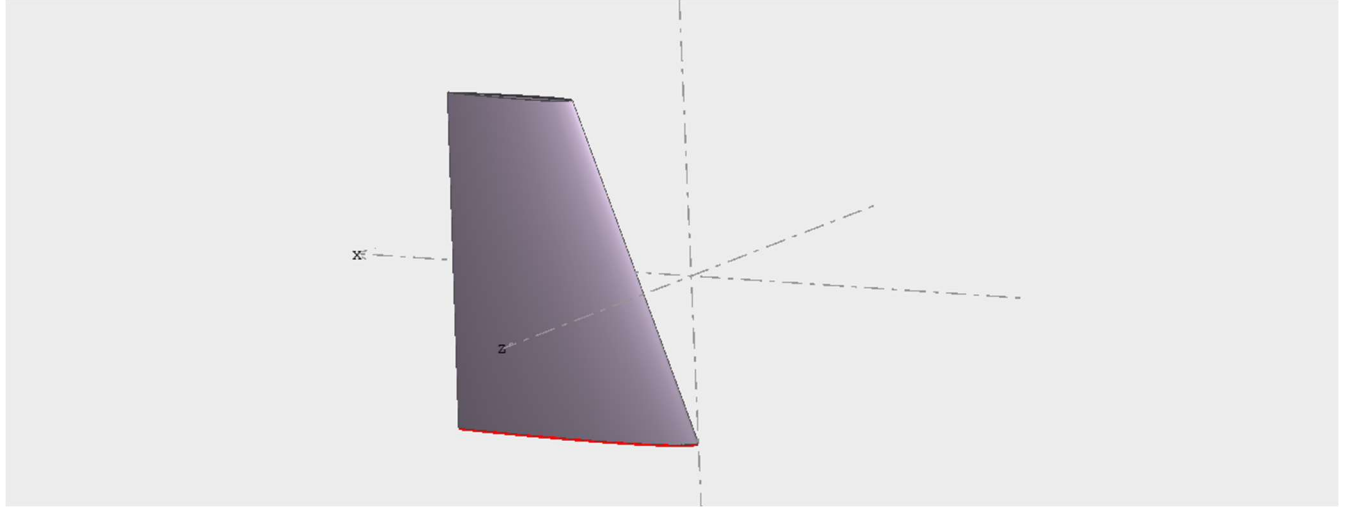


Figure 37 - 3D visual of the vertical tail (NACA 0012)

8 Static Stability Analysis

8.1 Longitudinal Stability

The longitudinal static stability of the aircraft was determined by calculating the coefficient of longitudinal stability, C_{M_α} . This coefficient represents the slope of the pitching moment curve with respect to angle of attack and indicates whether the aircraft will naturally return to equilibrium.

All the relevant input parameters used in the formula in Figure 38 are summarized in Table 12.

$$\begin{aligned}
 C_{M_\alpha} &= -\frac{dC_L}{d\alpha} \frac{x_w}{\bar{c}} + \frac{dC_D}{d\alpha} \frac{z}{\bar{c}} - \frac{dC_{L_{HT}}}{d\alpha} \frac{S_{HT}}{S\bar{c}} \eta_{HT} (l_{HT} + x_w) - \frac{d(C_{M_{cg}})_I}{d\alpha} \\
 &= -C_{L_\alpha} \frac{x_w}{\bar{c}} - \frac{dC_{L_{HT}}}{d\alpha_{HT}} \frac{d\alpha_{HT}}{d\alpha} \frac{S_{HT}}{S\bar{c}} \eta_{HT} (l_{HT} + x_w) - \frac{2\dot{m}l_I}{\rho V_\infty S\bar{c}} \frac{d\beta}{d\alpha} \\
 &= -C_{L_\alpha} \frac{x_w}{\bar{c}} - C_{L_{HT\alpha}} \left(1 - \frac{de}{d\alpha}\right) \left[C_{HT} + \frac{S_{HT}}{S\bar{c}} \eta_{HT} x_w\right] - \frac{2\dot{m}l_I}{\rho V_\infty S\bar{c}} \frac{d\beta}{d\alpha}
 \end{aligned}$$

Figure 38 - Longitudinal Stability Equation

Table 12 - Key Parameters Used in Longitudinal Stability Equation

Parameter	Value	Source/Method
Wing Lift Curve Slope	0.09	Calculated from $CL-\alpha$ curve
Distance CG to SC	0.25 m	Estimated as $\frac{1}{4}$ chord
Mean Chord Length	2.38 m	Geometric mean from wing
Downwash Rate	0.72	From graphs in course materials
Horizontal Tail Volume Coefficient	0.7	Design estimate
Horizontal Tail Area	18.88 m^2	CAD output
Horizontal Tail Efficiency	0.85	Standard design reference
Engine Mass Flow Rate	2.68 lb/s	Specification
Distance: Inlet to CoM	-0.74 m	Geometry estimate
CG to Tail Distance	-2.44 m	Measured from wing trailing edge
CG to inlet	3.0 m	Geometry estimate

Using these values, the final calculated value of the longitudinal stability coefficient is:

$$C_{M_\alpha} = -1.04$$

This result falls within the recommended stability range for general aviation aircraft, which is:

$$-1.5 < C_{M_\alpha} < -0.6$$

Thus, the aircraft demonstrates appropriate longitudinal static stability for safe and predictable flight performance. Additionally, this calculation was done using the formula sheet equation for C_{L_α} . This gave a C_{M_α} of -1.47 , which is still in the desired range.

8.2 Directional Stability

The directional static stability of the aircraft was determined by calculating the coefficient of directional stability, C_{n_β} . This coefficient represents the yawing moment response of the aircraft with respect to sideslip angle and indicates whether the aircraft will realign itself

with the relative wind in the presence of a yaw disturbance. All the relevant input parameters used in the formula in Figure 39 are summarized in Table 13.

$$C_{n\beta} = \frac{dC_n}{d\beta} = (C_{n\beta})_F + (C_{n\beta})_{VS} + (C_{n\beta})_W > 0.$$

Vertical stabilizer

$$(C_{n\beta})_{VS} = \bar{V}_{VS} (C_{L\alpha})_{VS} \left(1 + \frac{d\sigma}{d\beta}\right) \frac{qVS}{q}$$

$$\bar{V}_{VS} = \frac{l_{VS}S_{VS}}{bS}$$

$$\left(1 + \frac{d\sigma}{d\beta}\right) \frac{qVS}{q} = 0.724 + \frac{3.06S_{VS}/S}{1 + \cos \Lambda_{VS}} + 0.4 \frac{z_w}{h} + 0.009AR$$

Main wing

$$(C_{n\beta})_W = \frac{C_L^2}{4\pi AR} - \frac{C_L^2 \tan \Lambda}{\pi AR(AR + 4 \cos \Lambda)} \times$$

$$\left(\cos \Lambda - \frac{AR}{2} - \frac{AR^2}{8 \cos \Lambda} + 6 \frac{x \sin \Lambda}{\bar{c} AR} \right)$$

x – distance between c.g. and wing a.c.

Fuselage

$$(C_{n\beta})_F = -1.3 \frac{(VOL)_F}{Sb} \frac{h}{w},$$

$(VOL)_F$ - total volume of the fuselage
 h and w - average height and width of the fuselage.

Figure 39 – Directional Stability Equation

Table 13 - Key Parameters Used in Directional Stability Equation

Parameter	Value	Source/Method
Vertical Tail Area	2.535 m ²	CAD output
Distance CG to Vertical Tail	4.49 m	Measured from aircraft geometry
Vertical Tail Aspect Ratio	1.71	Geometry calculation
Distance from CG to Tailplane	2.0 m	Geometry calculation
Vertical Tail Lift Curve Slope	2.05	From standard NACA 0012 airfoil data
Vertical Tail Sweep	35°	Pre-determined for max stability
Interference Factor	1.05	Assumed for conventional tail configuration

Using these values, the final calculated value of the directional stability coefficient is:

$$C_{n_\beta} = 0.08$$

This result falls within the recommended stability range for general aviation aircraft, which is:

$$0.08 < C_{n_\beta} < 0.28$$

Thus, the aircraft demonstrates appropriate directional static stability for safe and predictable yaw behaviour.

8.3 Lateral Stability

The lateral static stability of the aircraft was approximated using the class method, which defines the lateral stability coefficient, C_{l_β} , as the negative of the directional stability coefficient, C_{n_β} . This is based on the principle that the vertical stabilizer contributes stabilizing effects to both yaw and roll, and their magnitudes are closely related.

$$C_{l_\beta} = -C_{n_\beta}$$

Given the previously calculated value of $C_{n_\beta} = 0.08$, the resulting lateral stability coefficient is:

$$C_{l_\beta} = -0.08$$

This result falls within the recommended range for general aviation aircraft lateral stability:

$$-0.08 < C_{l_\beta} < -0.28$$

Therefore, the aircraft is expected to display appropriate lateral static stability, contributing to balanced and unpredictable rolling motion in response to sideslip.

8.4 V-n Diagram

The V-n diagram, shown in Figure 40, represents the structural flight envelope of the aircraft and was generated based on calculated load factors at varying airspeeds. The blue lines outline the nominal flight envelope, showing the maximum and minimum allowable load factors under standard operating conditions. The orange lines represent the effect of wind gusts, using a maximum gust intensity of 66 ft/s and a dive condition gust of 25 ft/s. A safety factor of 1.5 was applied to these calculations to evaluate structural performance under more extreme conditions, and the resulting expanded envelope is shown in green.

As shown in the diagram, the impact of gust loads on the operational envelope becomes less significant when the safety factor is applied. This shows that the aircraft is structurally sound

to gust-induced loads within the expected flight regime, with the safety margin providing added assurance against unexpected loading.

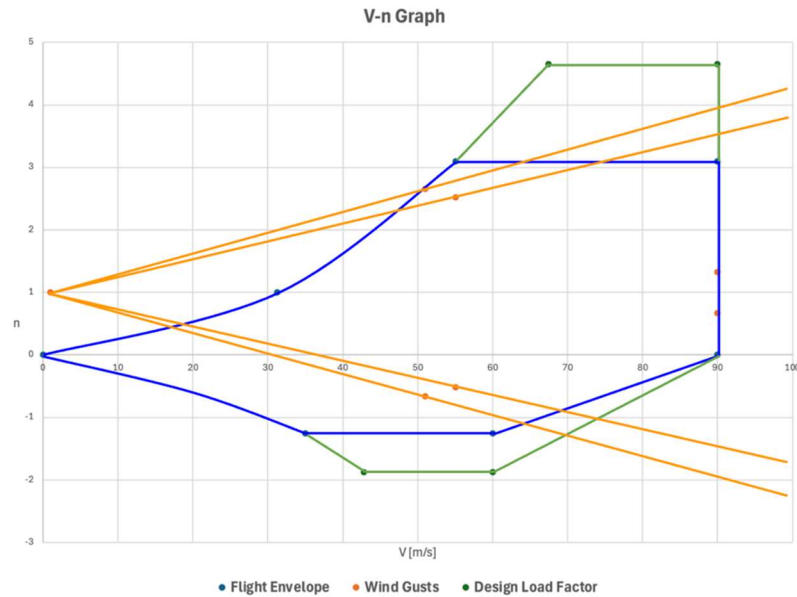


Figure 40 – V-n Diagram

9 Emissions and Cost Analysis

9.1 Noise and Pollutant Emissions

Inputting the aircraft's parameters, seen below in Table 14 relating to noise emissions, we were able to obtain a plot of SPL (dB) vs number of blades. For the 5 prop fans used in the landing configuration of the aircraft the max dB was 72.5dB from an observer 150 ft away. This is well within regulation from Annex 16, Volume I, Part II "Aircraft Noise Certification."

Table 14 - Parameters for noise calculation in MATLAB

Variable	Value
Harmonic Number	1
Number of Blades	5
Number of Rotors	6
Radius (m)	0.8
Solidity Ratio (%)	8
Max t/c	0.12
Mach Tip	0.755
Power Per Rotor (kW)	120
Thrust Per Rotor (N)	261.25
Altitude from Observer (ft)	150

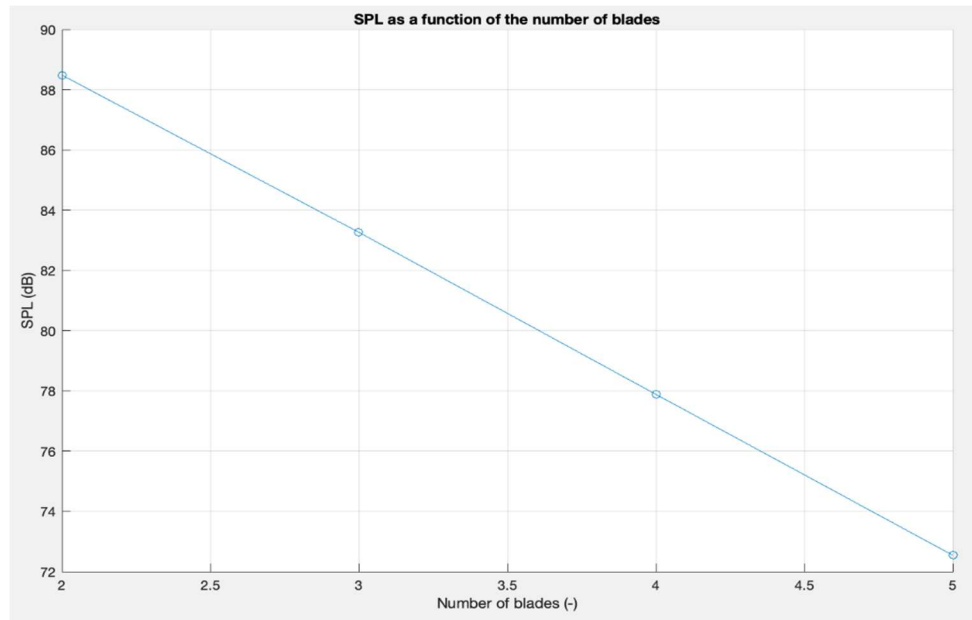


Figure 41 - SPL as a function of blades

The aircraft requires 60kg of Jet fuel and a 290kW.h battery to meet its designed flight time. From Lecture 17 - Pollutant and Noise Emissions, we can take the specific energy of Li-ion as 0.9[MJ/kg] and Jet Fuel A1 as 43.2[MJ/kg]. We then also take that the emissions from producing Li-ion as 87.6 [kgCO₂-eq/(kWh)] and for Jet Fuel A1 as 0.315 [kgCO₂-eq/(kWh)]. We also account for the emissions from burning Jet Fuel A1 assuming we burn the whole tank, and for the emissions generated by the Fort Nelson energy grid (the grid the batteries are charged on). The values are summarised below in Table 15.

Table 15 – Summary of Emissions

Jet Fuel A1	Emissions [kg CO ₂] per refill		
Capacity	Production	Burnt	Total
60kg	226.8	189.6	416.4
Li-ion Battery	Emissions [kg CO ₂] per cycle		
Capacity	Production	Per-Charge	Total
290 kWh	12.702	2.871	15.573
Battery Service Life	15 Years	~2000cycles	

Note for the above table the value for the batter's production is accounted for per cycle however in real life it would be 25404 [kgCO₂] in the year it's made then nothing during its service life.

9.2 Development and Production Costs

Developments costs for the aircraft are estimates based of historical records from [22]. Additional cost variables are estimated based on [23], which include research and design, manufacturing, tooling, custom designed motors. All components of cost are summarized below in Table 16. The estimated cost per unit produced would be 5.842[M\$] and price per unit sold would be 6.092[M\$]. This provides a profit margin 0.25[M\$].

Table 16 - Aircraft Cost Breakdown

DT&E/Production Costs (in Millions of \$)	Hour breakdown		Cost breakdown		
Type	DT&E	Production	DT&E	Production	TOTAL
Airframe Engineering	299659.69	209986.74	26.624	18.657	45.281
Manufacturing labour	130038.36	919671.83	9.801	69.314	79.115
Tooling	154274.15	209164.53	14.538	19.710	34.248
Quality Control					
Transport and Cargo Aircraft	9882.92	69895.06	0.818	5.787	6.606
Other Aircraft	16904.99	119557.34	1.400	9.899	11.299
Development support	--		14.2318896	--	14.2318896
Flight test operations			6.95006176		6.95006176
Manufacturing material and equipment			2.94446877	38.54437604	41.4888448
Engine			0.43809389	10.9523473	10.9523473
Avionics				2.25	125
Newly-built facilities					
Testing	--		0	--	0
Manufacturing			--	0	0
Profit			0	--	0
TOTAL					
Transport and Cargo Aircraft	--		78.596	287.965	366.561
Other Aircraft			79.177	292.077	371.254
UNIT COST					
Transport and Cargo Aircraft	--		5.759		-
Other Aircraft			5.842		
UNIT PRICE					
Transport and Cargo Aircraft	--			6.009	-
Other Aircraft				6.092	

10 CAD Model

The 3D model was created in SolidWorks from many parts and assemblies. Each component was modelled to the correct scale as specified throughout the project, reflecting the real-world scale of all components. This made sure that all components could realistically be built and assembled. Notable internal components in the fuselage include a pilot seat and a stretcher above the floor, and the turboshaft and battery below the floor, modeled to scale. Also note the door between wings hinges downward and contains a ramp for stretcher and crew while also being large enough for safety regulations.

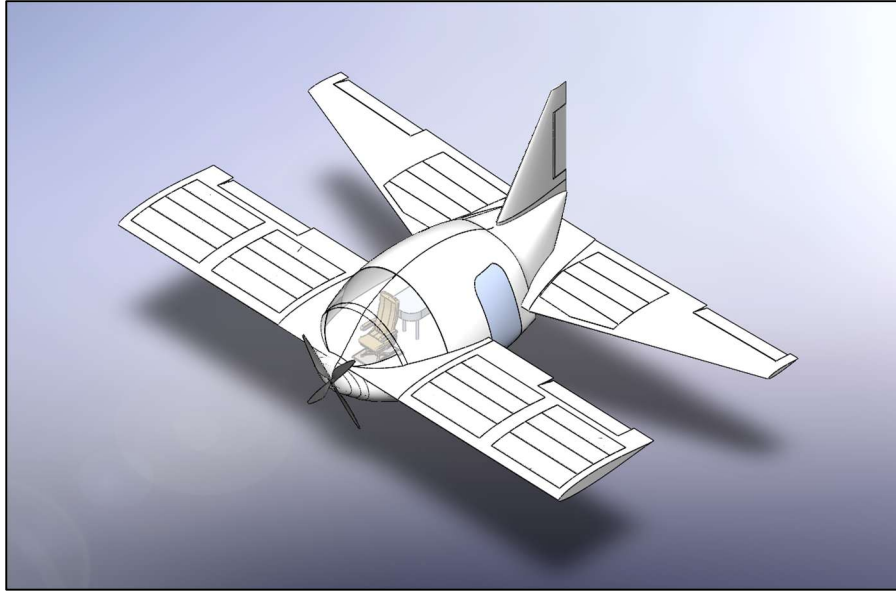


Figure 42 – Isometric with flaps closed

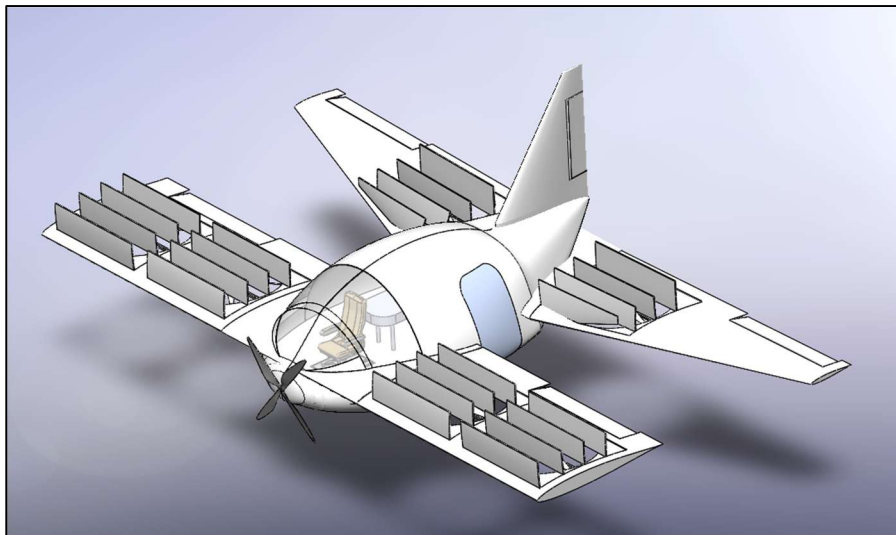


Figure 43 – Isometric with flaps open

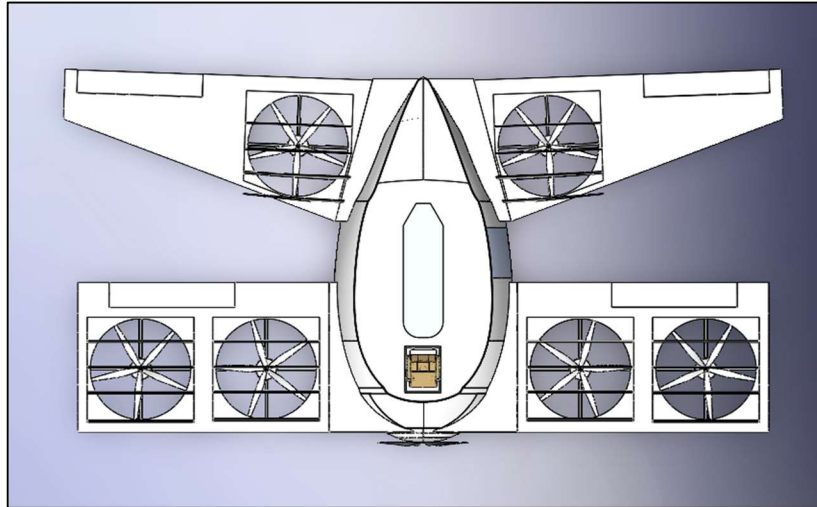


Figure 44 – Top down with flaps open

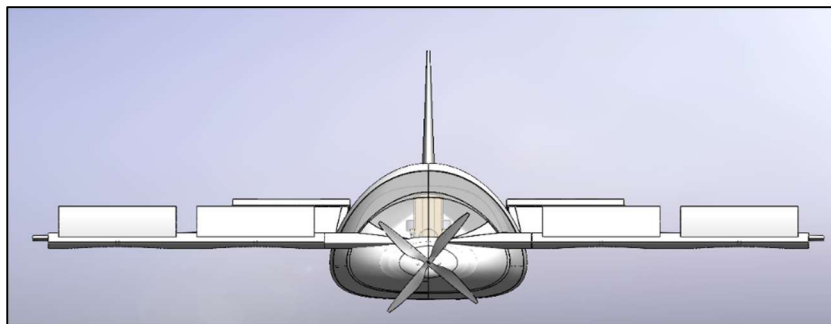


Figure 45 – Front view of aircraft

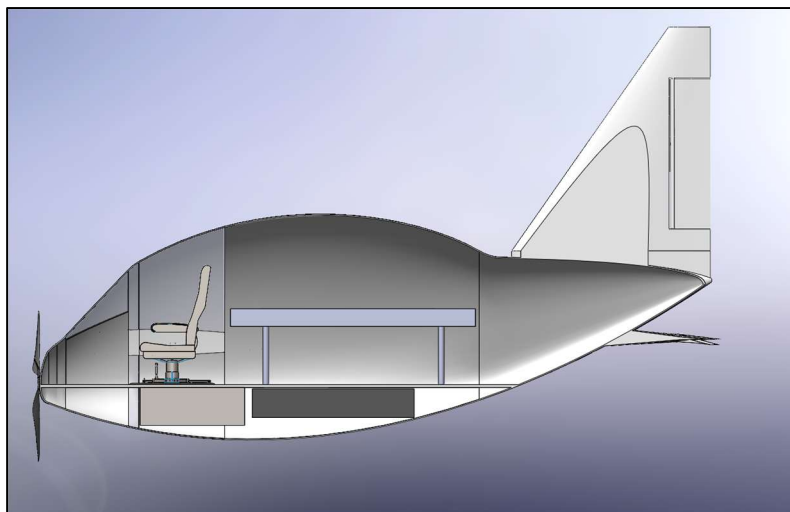


Figure 46 – Fuselage Cross Section

11 Conclusion and Next Steps

In several regions in northern British Columbia, emergency medical services face unique barriers in providing rapid and efficient response to accident scenes. This report provides an alternative design to typical ground and air ambulance vehicles with a focus on optimization for long-distance dispatch in areas with lower population density. A 200km round-trip mission profile was selected based on current market options, and six VTOL aircraft concept designs were proposed. A lift and cruise design with ducted fans being selected for further development.

The final concept features a main wing and tandem wing each with a span of 11m, with a rectangular main wing and a tapered tandem wing. Each half-span of the main wing houses two ducted fans driven by electric motors, while the tandem wing half-spans house a single ducted fan. The ducted fans are sized to provide enough lift to facilitate vertical take-off and landing. A turboprop located at the front of the fuselage is used for forward flight, and novel shutter coverings are featured over each fan to reduce drag during forward flight.

The aircraft has a maximum take-off mass (MTOM) of 2453.61 kg, with 54% being empty mass and 500kg being payload mass (4 passengers and 100kg of medical equipment). In vertical flight, the aircraft has a weight-to-disk-area ratio of 2093.98 N/m² and a weight-to-power ratio of 0.0076 N/W. In the forward flight, the aircraft has a wing loading of 560.52 N/m² and a weight-to-power ratio of 0.0252 N/W. Coefficients of longitudinal, directional and lateral stability were found to be within the acceptable range for general aviation aircrafts.

Looking forward, the next steps would involve further refinement of the design and the development of more detailed components. Most notably, the ducted fans and fan covering will require extensive research and design to ensure they can function as intended. Due to space limitations in the wings, the ducted fans drive system will require custom designed components, as will the fan covering system. From this point, detailed electrical and mechanical system layouts should be developed, outlining all required components.

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Appendix A – Criteria Ranking

Table A - 1 Speed criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift-Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	2/1	1/4	1/6	1/6	1/6
Fixed Rotor Quadcopter	1/2	1	1/7	1/8	1/8	1/8
Lift-Cruise Ducted Fans	4/1	7/1	1	2/1	1/3	3/1
Gyroplane	6/1	8/1	1/2	1	1	1
Tilt-rotor Micro Jet	6/1	8/1	3/1	1	1	1
Lift + Cruise	6/1	8/1	1/3	1	1	1

Table A - 2 Pollution criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift-Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	2/1	1/2	3/1	4/1	4/1
Fixed Rotor Quadcopter	1/2	1	3/1	4/1	5/1	6/1
Lift-Cruise Ducted Fans	1/2	1/3	1	3/1	4/1	3/1
Gyroplane	1/3	1/4	1/3	1	4/1	2/1

Tilt-rotor Micro Jet	1/4	1/5	1/4	1/3	1	2/1
Lift + Cruise	1/4	1/6	1/3	1/2	1/2	1

Table A - 3 Capacity criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift- Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	2/1	1/4	1/4	1/5	1/7
Fixed Rotor Quadcopter	1/2	1	1/5	1/5	1/6	1/8
Lift-Cruise Ducted Fans	4/1	5/1	1	1/3	1/2	1/4
Gyroplane	4/1	5/1	3/1	1	2/1	1/4
Tilt-rotor Micro Jet	5/1	6/1	2/1	1/2	1	1/3
Lift + Cruise	7/1	8/1	4/1	4/1	3/1	1

Table A - 4 Total area criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift- Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	1/2	4/1	4/1	5/1	7/1
Fixed Rotor Quadcopter	2/1	1	5/1	5/1	6/1	8/1
Lift-Cruise Ducted Fans	1/4	1/5	1	3/1	2/1	4/1

Gyroplane	1/4	1/5	1/3	1	1/2	4/1
Tilt-rotor Micro Jet	1/5	1/6	1/2	2/1	1	3/1
Lift + Cruise	1/7	1/8	1/4	1/4	1/3	1

Table A - 5 Noise criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift- Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	1/2	5/1	6/1	7/1	6/1
Fixed Rotor Quadcopter	2/1	1	4/1	4/1	5/1	4/1
Lift-Cruise Ducted Fans	1/5	1/4	1	3/1	2/1	4/1
Gyroplane	1/6	1/4	1/3	1	1/2	4/1
Tilt-rotor Micro Jet	1/7	1/5	1/2	2/1	1	3/1
Lift + Cruise	1/6	1/4	4/1	1/4	1/3	1

Table A - 6 Cost criteria ranking

Alternative	Tilt-Rotor Quadcopter	Fixed Rotor Quadcopter	Lift- Cruise Ducted Fans	Gyroplane	Tilt-rotor Micro Jet	Lift + Cruise
Tilt-Rotor Quadcopter	1	1/3	5/1	4/1	3/1	5/1
Fixed Rotor Quadcopter	3/1	1	6/1	5/1	6/1	8/1

Lift-Cruise Ducted Fans	1/5	1/6	1	3/1	4/1	2/1
Gyroplane	1/4	1/5	1/3	1	1/2	5/1
Tilt-rotor Micro Jet	1/3	1/6	1/4	2/1	1	4/1
Lift + Cruise	1/5	1/8	1/2	1/5	1/4	1

Appendix B – Engine & Motor Sizing Reference Values

Table B - 7 Engine/Propeller Sizing & Specifications

<i>Name</i>	<i>Variable</i>	<i>Value</i>	<i>Units</i>	<i>Source</i>
Main Wing Aspect Ratio	AR	4.620	[-]	Chosen
Zero Lift Drag	C_{D_0}	0.012	[-]	Calculated
Lift Coefficient	C_L	0.546	[-]	Calculated
Propeller Power Coefficient	C_P	0.129	[-]	Calculated
Propeller Thrust Coefficient	C_T	0.112	[-]	Graph
Propeller Diameter	D_{prop}	2.000	[m]	Chosen
Efficiency Factor	e	0.800	[-]	Standard
Propeller Efficiency (Graph)	η_P	0.870	[-]	Graph
Propeller Efficiency (Calculated)	η_P	0.870	[-]	Calculated
Propeller Advance Ratio	J	1.000	[-]	Calculated
Propeller Tip Mach Number	M_{tip}	0.800	[-]	Standard
Maximum Take Off Mass	MTOM	2400	[kg]	Chosen
Propeller Rotational Speed	n	30.000	[rev/s]	Standard
Number of Propeller Blades	NB_{prop}	4.000	[-]	Chosen
Sized Engine Power	P_{eng}	112.000	[kW]	Chosen
Required Power in Cruise	P_{engreq}	111.938	[kW]	Calculated
Air Density at Altitude	ρ_∞	1.007	[kg/m ³]	Condition
Propeller Disc Area	S_{prop}	3.142	[m ²]	Calculated
Wing Reference Area	S_{wing}	23.810	[m ²]	Chosen

Engine Thrust	T_{eng}	1624.000	[N]	Calculated
Required Thrust in Cruise	T_{req}	1623.108	[N]	Calculated
Airflow Velocity	V_{∞}	60.000	[m/s]	Condition
Tip Speed of Propeller Blade	$V_{tipprop}$	197.814	[m/s]	Calculated

Table B - 8 Electric Ducted Fan/Rotor Sizing & Specifications

<i>Name</i>	<i>Variable</i>	<i>Value</i>	<i>Units</i>	<i>Source</i>
Base Drag Coefficient	$C_{D_{0fan}}$	0.01	[-]	Chosen
Fan Diameter	D_{fan}	1.6	[m]	Chosen
Rotor Efficiency	η_R	85	[%]	Chosen
Electric Motor Efficiency	η_{motor}	95	[%]	Chosen
Maximum Take Off Mass	MTOM	2400	[kg]	Chosen
Fan Baldes per Fan	NB_{fan}	5	[-]	Chosen
Rotor Speed	n_{rot}	49.74	[rev/s]	Calculated
Required Electric Motor Power	$P_{motorreq}$	167.08	[kW]	Calculated
Sized Electric Motor Power	P_{motor}	170	[kW]	Chosen
Total Motor Power Required	P_{fan}	1002.46	[kW]	Calculated
Hover Power to Weight	P/W_h	32.691	[W/N]	Calculated
Vertical Climb Power to Weight	P/W_{vc}	34.382	[W/N]	Calculated
Air Density at Altitude	ρ_{∞}	1.058	[-]	Chosen
Rotor Solidity	σ	8	[%]	Chosen
Total Fans	TF	6	[-]	Chosen
Induced Velocity in Hover	$V_{i,h}$	26.301	[-]	Calculated

Fan Tip Velocity	V_{tipfan}	250	[m/s]	Chosen
Fan Loading	W/A	1463.73	[N/m ²]	Calculated
Needed Fan Power Loading	W/P	0.02909	[N/W]	Calculated
Climb Velocity	V_v	4	[m/s]	Chosen

Appendix C – Static Stability Reference Values

Table C - 9 Reference Values for Longitudinal Stability

<i>Name</i>	<i>Variable</i>	<i>Value</i>	<i>Units</i>	<i>Source</i>
Change of Lift Coefficient w.r.t Angle of Attack of Wing from XLFR5	$C_{L\alpha}$	0.09	[-]	Calculated
Change of Lift Coefficient w.r.t Angle of Attack of Wing Calculated from Formula Sheet (reference value)	$C_{L\alpha}$	4.1654	[-]	Calculated
Dist. from Aircraft CG to Wing AC	X_W	0.25	[m]	Calculated
Wing Chord Length	$\bar{c}$	2.38	[m]	Calculated
Angle of Attack Coefficient of Lift of Tandem Wing	$C_{L\alpha HT}$	4.6636	[-]	Calculated
Tandem Wing Taper Ratio	Λ	0.3	[-]	Chosen
Tandem Wing Aspect Ratio	AR	6.41	[-]	Calculated
Tandem Wing Span	b	11	[m]	Chosen
Horiz. Dist. Between a.c. of Wings	L	2.85	[m]	Chosen
Vertical Distance Between Wings	H	0	[m]	Chosen
Change of Downwash Angle w.r.t. Angle of Attack	$\frac{de}{d\alpha}$	0.72	[-]	Calculated
Horizontal Tail Coefficient	C_{HT}	0.7	[-]	Standard
Tandem Wing Area	S_{HT}	18.88	[m ²]	Calculated
Horizontal Tail Efficiency	η_{HT}	0.85	[-]	Estimate
Engine Mass-Flow Rate	$\dot{m}$	2.68	[kg/s]	Estimate
Distance from Engine Inlet to CoM	L_i	-0.74	[m]	Chosen
Air Density @ Cruise (1500m)	ρ	1.007	[kg/m ³]	Standard

Coefficient of Longitudinal Stability	$C_{M\alpha}$	-1.4683	[-]	Calculated
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Table C - 10 Reference Values for Directional Stability

<i>Name</i>	<i>Variable</i>	<i>Value</i>	<i>Units</i>	<i>Source</i>
Average Fuselage Height	h	1	[m]	Calculated
Average Fuselage Width	w	2.3	[m]	Calculated
Total Fuselage Volume	VOL_f	27	[m ³]	Calculated
Wing Area	S	52.38	[m ²]	Calculated
Wing Span	b	11	[m]	Chosen
Fuselage Directional Stability	$(C_{n_\beta})_F$	-0.029	[-]	Calculated
Dist. Between c.g. and stabilizer a.c	l_{vs}	4.49	[m]	Calculated
Vertical Tail Area	S_{VT}	2.535	[m ²]	Calculated
Tail Aspect Ratio	AR_{VT}	1.5	[-]	Chosen
Tail Height	h_{VT}	1.95	[m]	Chosen
Vertical Offset Distance of Main Wing to Fuselage	Z_W	0.25	[m]	Chosen
Leading Edge Sweep of Tail	Λ_{VT}	35	[°]	Chosen
Vertical Stabilizer Directional Stability	$(C_{n_\beta})_{VS}$	0.0944	[-]	Calculated
Square of Main Wing Lift Coef.	$(C_L)^2$	1	[-]	Standard
Main Wing Sweep Angle	Λ	0	[°]	Chosen
Main Wing Aspect Ratio	AR	4.62	[-]	Calculated
Main Wing Average Chord Length	$\bar{c}$	2.38	[m]	Calculated
Main Wing Directional Stability	$(C_{n_\beta})_W$	0.01722	[-]	Calculated
Total Directional Stability	C_{n_β}	0.0825	[-]	Calculated

Appendix D – Fuselage Drag Estimation Reference Values

Table D - 11 Reference Values for Fuselage Drag Calculation

Name	Variable	Value	Units	Source
Cruise Velocity	V_{cr}	60	[m/s]	Chosen
Mach at Cruise	M_{cr}	0.174927	[-]	Calculated
Wing Span	b	11	[m]	Chosen
Wing Area	A	26.19	[m ²]	Calculated
Aspect Ratio	AR	4.62	[-]	Chosen
Wing Reference Area	S_{wing}	26.19	m ²	Calculated
Density	ρ	1.007	kg/m ³	Standard
Horizontal Component of Forward Flight Velocity	V_{∞}	60	m/s	Chosen
Lift-to-Drag Ratio	l/d	20.5	[-]	Simulated
Form Factor	F	1.058	[-]	Calculated
Interference Factor	Q	1	[-]	Standard
Top View Projected Area of Nose	A_{top}	9.66	m ²	Calculated
Side View Projected Area of Nose	A_{side}	5.8	m ²	Calculated
Wetted Area	S_{wet}	24.284	m ²	Calculated
Surface Type	k	4.05E-06	m	Metal Shell
Length of Nose	l	0.14	m	Chosen
Reynolds Number	Re	2.30E+06	[-]	Calculated
Dynamic Viscosity	μ_{cr}	1.40E-05	Pa.s	Standard

Reynolds Number w/o Effect of Surface Roughness	Re	6.04E+05	[-]	Calculated
Worst Case Scenario Reynolds Number	Re	604200 (laminar)	[-]	Calculated
Laminar Coefficient at Nose	C_f	0.001981827	[-]	Calculated
Turbulent Coefficient at Nose	C_f	0.013574917	[-]	Calculated
Total Friction Force Along Nose	F_f	92.31467654	N	Calculated
Top View Projected Area of Fuselage	A_{top}	11.63	m ²	Calculated
Side View Projected Area of Fuselage	A_{side}	7.419	m ²	Calculated
Wetted Area	S_{wet}	29.922	m ²	Calculated
Surface Type	k	4.05E-06	m	Metal Shell
Length of Fuselage	l	5.74	m	Chosen
Reynolds Number	Re	1.15E+08	[-]	Calculated
Dynamic Viscosity	μ_{cr}	1.40E-05	Pa.s	Standard
Reynolds Number w/o Effect of Surface Roughness	Re	2.48E+07	[-]	Calculated
Worst Case Scenario Reynolds Number	Re	24772200 (turbulent)	[-]	Calculated
Laminar Coefficient at Nose	C_f	0.000309509	[-]	Calculated
Turbulent Coefficient at Nose	C_f	0.008298804	[-]	Calculated
Total Friction Force Along Nose	F_f	476.3054509	N	Calculated
Total Friction acting on Fuselage	$F_{f, total}$	568.6201274	N	Calculated
Base Drag Coefficient	C_{D0}	0.011977795	[-]	Calculated

Appendix E – Structural Reference Values (V-n Diagram)

Table E - 12 Reference Values for V-n Diagram

Name	Variable	Value	Units	Source
Cruise Velocity	V_{cr}	60	[m/s]	Chosen
Maximum Take Off Mass	MTOM	2453.61	[kg]	Calculated
Main Wing Area	S_W	26.19	[m ²]	Calculated
Wing Span	b	11	[m]	Chosen
Aspect Ratio	AR	4.62	[-]	Calculated
Maximum Coefficient of Lift	$C_{L_{max}}$	1.5	[-]	Standard
Dive Speed	V_D	90	[m/s]	Calculated
Positive Limit Load	n_+	3.1	[-]	Standard
Negative Limit Load	n_-	-1.25	[-]	Standard
Density at Sea Level	ρ_{SL}	1.225	[kg/m ³]	Standard
Density at Cruise Level	ρ	1.058	[kg/m ³]	Standard
Limit Load, Velocity	$n_{velocity}$	1	[-]	Standard
Stall Velocity	V_s	31.63	[m/s]	Calculated
Limit Load, Manoeuvring Speed	$n_{manoeuvring}$	3.1	[-]	Standard
Design Maneuvering Speed	V_a	55.69	[m/s]	Calculated

Table E - 13 Reference Values for Gust Calculations on V-n Diagram

Name	Variable	Value	Units	Source
Root Chord	c_{root}	2.38	[m]	Calculated
Mass Ratio	μ	17.86	[m]	Calculated

Design Factor	K	0.682	[-]	Calculated
Wing Loading	W/S	919.03	[N/m ²]	Calculator
Flight Condition Gust, Max	$u_{de,max}$	66	[ft/s]	Standard
Flight Condition Gust, Level Flight	$u_{de,level}$	50	[ft/s]	Standard
Flight Condition Gust, Dive	$u_{de,dive}$	25	[ft/s]	Standard
Change in Load Factor, Cruise	Δn_{cruise}	1.49	[-]	Calculated
Change in Load Factor, Max Speed	Δn_{max}	0.32	[-]	Calculated
New Load Factor, Cruise	n_{cruise}	2.49	[-]	Calculated
New Load Factor, Max Speed	n_{max}	1.32	[-]	Calculated
Min. Velocity for Max Gust	V_b	49.95	[m/s]	Calculated
Change in Load Factor, Max Gust	$\Delta n_{max\ gust}$	1.64	[-]	Calculated
New Load Factor, Max Gust	$n_{max\ gust}$	2.64	[-]	Calculated

Table E - 14 Reference Values for Design Load Factor Calculations on V-n Diagram

<i>Name</i>	<i>Variable</i>	<i>Value</i>	<i>Units</i>	<i>Source</i>
Factor of Safety	FS	1.5	[-]	Standard
Design Load Factor, Positive	$n_{design+}$	4.65	[-]	Calculated
Design Load Factor, Negative	$n_{design-}$	-1.875	[-]	Calculated
Velocity, Positive Load Factor	$V_{design+}$	68.20	[m/s]	Calculated
Velocity, Negative Load Factor	$V_{design-}$	43.31	[m/s]	Calculated